CFA LEVEL I

数量分析
Quantitative Methods

中英文原题速学讲义

A bilingual, concept-first guide to the supplied questions

93 道原题 93 original questions	9 个学习单元 9 study units	20 道核心题 20 core examples

先掌握知识点，再用原题验证。

Learn the concept, work the original example, then check the trap.

包含：完整英文原题、中文翻译、答案、双语解析、公式速查、答案总表与原文勘误。

Includes every original English question, Chinese translation, answers, bilingual reasoning, formulas, answer index and source notes.

资料范围：题库数量分析部分，第 1-93 题。教材年份及原始官方/模拟出处未标明；本讲义的单元编号为学习重组。

Scope: Quantitative Methods Q1-Q93 in the supplied bank. Curriculum year and original official/mock provenance are not identified. Unit numbers are editorial study groupings.

整理日期 / Prepared: 2026-11-02

如何最快学会

How to learn efficiently

每个小节都按同一顺序阅读：知识点 → 公式及适用条件 → 原题 → 中文翻译 → 逐步解析 → 易错点。

Follow the same sequence in each section: concept → formula and conditions → original question → Chinese translation → worked solution → common trap.

90 分钟第一轮 / A 90-minute first pass

前 10 分钟浏览各单元知识点；接着用约 60 分钟完成下列 20 道核心题；最后 20 分钟只复盘错误、猜对及超时题。时间为建议，可按基础调整。

Spend 10 minutes scanning the concepts, about 60 minutes on the 20 core questions below, and 20 minutes reviewing incorrect, guessed or slow answers. Adjust the timing to your starting level.

学习顺序 / Sequence	核心原题 / Core originals
收益率 / Returns	Q03 · Q06 · Q37 · Q54
货币时间价值 / Time value	Q13 · Q63
统计与概率 / Statistics and probability	Q45 · Q55 · Q52 · Q57
组合风险 / Portfolio risk	Q62
抽样与检验 / Sampling and testing	Q01 · Q51 · Q93 · Q17 · Q83
回归 / Regression	Q02 · Q23
金融科技 / Fintech	Q12 · Q77

第二轮完成全部 93 题。重复考点用于巩固，不增加自编题；每道题保留原编号，PDF 书签可直接跳转。

For the second pass, complete all 93 questions. Repeated concepts provide reinforcement; no new questions have been added. Original IDs are retained and PDF bookmarks provide direct navigation.

错题复盘 / Mistake review

把错误归为：概念混淆、公式选择、现金流时间点、运算或单位。猜对的题也要复盘；重做相应小节，直到能说清为什么其他选项不对。

Classify mistakes as concept confusion, formula selection, cash-flow timing, or arithmetic and units. Review guessed-correct answers too, and rework the relevant section until you can explain why the other choices fail.

资料与阅读说明

Sources and reading conventions

原题来源：1000题(9).pdf，PDF 第 115-133 页，数量分析章节印刷页码 1-19。答案来源：数量分析(2).pdf，第 1-93 页；第 n 页对应原题 Qn。

Question source: 1000题(9).pdf, PDF pages 115-133, Quantitative Methods printed pages 1-19. Answer source: 数量分析(2).pdf, pages 1-93; page n corresponds to Qn.

英文原题以原 PDF 的矢量内容保留，含原表格、数字及选项；中文和解析为本讲义整理。答案不是简单照抄：计算题已复算，原文的缺项或错印另行标注。

The original English questions are retained as vector PDF content, including tables, numbers and choices. Chinese translations and teaching explanations are editorial additions. Numerical solutions were recalculated, and source omissions or misprints are marked separately.

标记 / Label	含义 / Meaning
核心例题 / Core example	第一轮优先学习的 20 题。 The 20 questions prioritized for the first pass.
巩固练习 / Reinforcement	完整保留的其余原题，按同类考点排列。 The remaining original questions, grouped by concept.
资料提示 / Source note	原题或答案存在缺项、错印，或需要说明适用范围。 An omission, misprint or wording qualification in the supplied material.
原答案 / Supplied answer	指第二份文件给出的答案，不表示已核实原始官方出处。 The answer in the supplied second file; original official provenance is unverified.

两处主要影响作答的原文问题：Q19 的题干表格不完整，本讲义只在解析中标明答案页提供的补充数据；Q74 的印刷选项与计算结果不符，保留原选项并区分“原答案 C”和正确计算值。

Two source issues materially affect answering: Q19 has an incomplete question table, so additional values from the solution page are labeled only in the explanation. Q74 has printed choices inconsistent with the calculation, so the original choices, supplied C key and computed value are distinguished.

教材年份未注明。本讲义覆盖你提供的原题范围，不把这些题自动标作某一年度完整考纲；金融科技等内容也按原题保留。

The curriculum year is not stated. This guide covers the supplied questions without presenting them as a complete syllabus for any particular year; fintech questions are retained as supplied.

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目录和核心题编号可点击；书签还包含全部 93 道题。 / Contents and core IDs are clickable; bookmarks also include every question.

单元 01 利率与收益率

Rates and Returns

先辨认题目在问什么：利率补偿、单期投资回报、复利年化，还是资金流影响。收益率题最常见的失分原因不是公式难，而是漏掉股息、错看时间或选错收益率口径。

First identify what is being measured: compensation through interest rates, a single-period return, compound annual growth, or the effect of cash flows. The main traps are omitted dividends, incorrect timing, and choosing the wrong return measure.

本单元 15 题 / 15 questions in this unit

单元速记 / Unit essentials

看到股息，先写“价格变化+股息”；看到年化，先写“财富倍数取幂”。

For dividends, start with price change plus income; for annualization, start with a wealth factor raised to a power.

评价投资者看MWR，评价不控制客户资金流的经理看TWR；始终先画时间点。

Use MWR for the investor and TWR for a manager who does not control client cash flows; establish the timing first.

1.1 实际利率、名义利率与机会成本 / Real rates, nominal rates, and opportunity cost

先学知识点 / Learn the concept first

实际无风险利率补偿放弃当前实际消费的时间偏好；通胀溢价补偿未来购买力下降。名义无风险利率在本题口径下约等于两者之和。

The real risk-free rate reflects the preference for current over future real consumption. The inflation premium compensates for expected loss of purchasing power. Under the convention used in these questions, their sum approximates the nominal risk-free rate.

比较机会成本时先确定参照方案。第10题问选择低息存款证相对最佳同风险替代品少赚多少；第25题问完全不投资而放弃的名义收益。

Identify the alternative before computing opportunity cost. Q10 asks for the return forgone relative to the best alternative with the same risk and maturity; Q25 asks for the nominal return forgone by not investing at all.

> Nominal risk-free rate ≈ Real risk-free rate + Inflation premium
>
> 名义无风险利率约等于实际无风险利率加通胀溢价。
>
> The nominal risk-free rate is approximately the real risk-free rate plus the inflation premium.

Q69 实际消费的时间偏好 / Time preference for real consumption

巩固练习 / Reinforcement • 原题 PDF p.129 / 章节 p.15 • 答案 PDF p.69

69. The time preference of individuals for current instead of future real consumption is captured by the:
    A. liquidity premium.
    B. maturity premium.
    C. real risk-free interest rate.

中文题干与选项 / Chinese translation

个人偏好当前而非未来实际消费的时间偏好，体现于：

A. 流动性溢价。

B. 期限溢价。

C. 实际无风险利率。

答案 / Answer: C

1. 实际无风险利率是在不考虑通胀和风险补偿时，推迟实际消费所要求的回报。

The real risk-free rate is the return required to postpone real consumption, apart from inflation and risk compensation.

2. 流动性溢价与变现难度有关，期限溢价与较长期限相关，均不是本题的消费时间偏好。

The liquidity premium concerns difficulty of converting an investment to cash, while the maturity premium concerns longer maturities; neither is the consumption preference described.

易错点 / Common trap

题干出现real consumption和current versus future，优先定位实际无风险利率。

The phrases real consumption and current versus future point to the real risk-free rate.

Q25 名义无风险利率 / Nominal risk-free rate

巩固练习 / Reinforcement • 原题 PDF p.119 / 章节 p.5 • 答案 PDF p.25

25. An investor is considering buying a US T-bill. If the real risk-free rate is 1% and the inflation premium is 2%, the investor's opportunity cost of failing to make this investment is *closest* to:
    A. 1%.
    B. 2%.
    C. 3%.

中文题干与选项 / Chinese translation

某投资者考虑购买美国短期国库券。如果实际无风险利率为1%，通胀溢价为2%，不进行这项投资的机会成本最接近：

A. 1%。

B. 2%。

C. 3%。

答案 / Answer: C

1. 短期国库券收益在题目中代表名义无风险收益，应同时包含实际无风险利率和通胀补偿。

The T-bill return represents a nominal risk-free return in this question, incorporating both the real risk-free rate and inflation compensation.

2. 按题库的加总口径：1% + 2% = 3%，选C。

Using the source's additive convention: 1% + 2% = 3%, so C is correct.

易错点 / Common trap

1%只反映实际回报，2%只反映通胀补偿；都不是放弃的完整名义收益。

The 1% covers only the real return, while 2% covers only inflation compensation; neither is the full nominal return forgone.

Q10 相对最佳替代品的机会成本 / Opportunity cost relative to the best alternative

巩固练习 / Reinforcement • 原题 PDF p.116 / 章节 p.2 • 答案 PDF p.10

10. An investor considers the following certificates of deposit (CDs) available for purchase at face value:

CD	Interest Rate
1	2.2%
2	3.3%
3	4.4%

If each CD has the same maturity and default risk, the opportunity cost of investing in CD 1 is *closest* to:

A. 0.0%.

B. 1.1%.

C. 2.2%.

中文题干与选项 / Chinese translation

某投资者考虑按面值购买下表三种存款证。若到期日和违约风险都相同，投资于存款证1的机会成本最接近：

存款证	利率
1	2.2%
2	3.3%
3	4.4%

A. 0.0%。

B. 1.1%。

C. 2.2%。

答案 / Answer: C

1. 相同到期日和违约风险下，最佳替代品是利率最高的存款证3，收益率为4.4%。

With equal maturity and default risk, the best alternative is CD 3, which offers 4.4%.

2. 按本题的比较口径，选择存款证1少赚4.4% - 2.2% = 2.2个百分点，选C。

Under the comparison used in this question, choosing CD 1 forgoes 4.4% - 2.2% = 2.2 percentage points, so C is correct.

易错点 / Common trap

应与最佳可行替代品比较，而不是与存款证2或三个利率的平均值比较。

Compare with the best available alternative, not CD 2 or the average of all three rates.

1.2 持有期收益：价格变化加现金收入 / Holding-period return: price change plus income

先学知识点 / Learn the concept first

持有期收益必须同时计算资本利得和持有期间收到的现金收入，并除以期初投入。题目说“收到股息后立即卖出”，股息仍然属于投资回报。

A holding-period return includes both the capital gain and cash income received, divided by the initial investment. Selling immediately after a dividend does not remove that dividend from the return.

反求卖出价格时，把已收到的股息从期末总财富中扣除；期末总财富与卖出价格不是同一个数。

When solving for the sale price, subtract the dividend already received from ending total wealth. Ending total wealth and the sale price are different quantities.

> HPR = (P1 - P0 + D)/P0; P1 = P0(1 + HPR) - D
>
> P0为买入价，P1为卖出价，D为期间现金收入。
>
> P0 is the purchase price, P1 the sale price, and D the cash income during the period.

Q03 持有期收益 / Holding-period return

核心例题 / Core example · 原题 PDF p.115 / 章节 p.1 · 答案 PDF p.3

3. An investor purchases a stock for $100. Immediately after receiving a dividend of $7, the investor sells the stock for $107. The holding period return of the investment is *closest* to:
   A. 0%.
   B. 7%.
   C. 14%.

中文题干与选项 / Chinese translation

某投资者以100美元买入一只股票，在收到7美元股息后立即以107美元卖出。该投资的持有期收益率最接近：

A. 0%。

B. 7%。

C. 14%。

答案 / Answer: C

1. 把价格收益与股息合并：资本利得为107 - 100 = 7美元，股息为7美元，总收益为14美元。

Combine the capital gain and dividend: 107 - 100 = $7 of capital gain plus $7 of dividends gives a total gain of $14.

2. HPR = (107 - 100 + 7)/100 = 14%，所以选C。

HPR = (107 - 100 + 7)/100 = 14%, so C is correct.

易错点 / Common trap

只算价格涨幅会误选7%；收到股息后卖出，不代表要从收益中扣掉股息。

Using only price appreciation gives the incorrect 7%; selling after receiving a dividend does not mean subtracting it from return.

Q09 由总收益反求卖出价 / Solving for the sale price from total return

巩固练习 / Reinforcement • 原题 PDF p.116 / 章节 p.2 • 答案 PDF p.9

9. An investor purchased a stock for $450 and then sold the stock immediately after receiving a dividend of $2. If the holding period return is a loss of 10.2%, the investor sold the stock at a price *closest* to:
   A. $402.
   B. $404.
   C. $406.

中文题干与选项 / Chinese translation

某投资者以450美元买入股票，在收到2美元股息后立即卖出。若持有期收益率为亏损10.2%，卖出价最接近：

A. 402美元。

B. 404美元。

C. 406美元。

答案 / Answer: A

1. 先写总回报：-0.102 = (P1 - 450 + 2)/450。

Start with total return: -0.102 = (P1 - 450 + 2)/450.

2. P1 = 450(1 - 0.102) - 2 = 402.10美元，最接近402美元，选A。

P1 = 450(1 - 0.102) - 2 = $402.10, closest to $402, so A is correct.

易错点 / Common trap

404.10美元是包含已收到股息的期末总财富；股息已经收到，股票卖价要再减2美元。

$404.10 is ending total wealth including the dividend already received; subtract $2 to obtain the stock sale price.

1.3 复利年化与复合增长率 / Compound annualization and compound growth

先学知识点 / Learn the concept first

先把收益率转换为财富倍数1+R，再按时间比例取幂，最后减1。多期累计收益不能直接除以年数得到复合年收益率。

Convert the return to a wealth factor, 1 + R, raise it to the appropriate time exponent, and subtract 1. Dividing a cumulative return by the number of years does not produce its compound annual rate.

若持有期用月表示，年化指数是12/月数；若题目给未来收入是今天的多少倍，直接对这个倍数开N次方。

For a holding period measured in months, the annualization exponent is 12 divided by the number of months. For a future-to-current revenue multiple, take the Nth root of that multiple.

R_annual = (1 + R_period)^(12/months) - 1

按月给定持有期时的复利年化公式。

Compound annualization when the holding period is specified in months.

CAGR = (FV/PV)^(1/N) - 1

N为年数，FV/PV为累计增长倍数。

N is the number of years and FV/PV is the cumulative growth multiple.

Q06 非整年收益年化 / Annualizing a partial-year holding period

核心例题 / Core example · 原题 PDF p.116 / 章节 p.2 · 答案 PDF p.6

6. An asset earns 13.1% over a 16-month period. The asset's annualized compound rate of return is *closest* to:
   A. 9.3%.
   B. 9.7%.
   C. 9.8%.

中文题干与选项 / Chinese translation

某资产在16个月内取得13.1%的收益。其年化复合收益率最接近：

A. 9.3%。

B. 9.7%。

C. 9.8%。

答案 / Answer: B

1. 16个月对应16/12年。年化时对财富倍数1.131取12/16次方。

Sixteen months equals 16/12 years. Annualize by raising the wealth factor 1.131 to the power 12/16.

2. R_annual = 1.131^(12/16) - 1 = 9.6723%，最接近9.7%，选B。

R_annual = 1.131^(12/16) - 1 = 9.6723%, closest to 9.7%, so B is correct.

易错点 / Common trap

13.1% × 12/16 = 9.825%是按比例缩放，不是复利年化；“每月复合收益×12”也不是有效年收益。

13.1% × 12/16 = 9.825% is linear scaling, not compound annualization. Multiplying a compound monthly return by twelve also does not give an effective annual return.

Q08 复合年增长率 / Compound annual growth rate

巩固练习 / Reinforcement · 原题 PDF p.116 / 章节 p.2 · 答案 PDF p.8

8. A company estimates its revenue will be 50% higher than today in four years' time. The compound annual growth rate is *closest* to:
   A. 10.7%.
   B. 11.8%.
   C. 12.5%.

中文题干与选项 / Chinese translation

一家公司预计四年后的收入比今天高50%。复合年增长率最接近：

A. 10.7%。

B. 11.8%。

C. 12.5%。

答案 / Answer: A

1. 四年后的收入为今天的1.50倍，因此(1 + g)^4 = 1.50。

Revenue in four years is 1.50 times current revenue, so (1 + g)^4 = 1.50.

2. g = 1.50^(1/4) - 1 = 10.6682%，选A。

g = 1.50^(1/4) - 1 = 10.6682%, so A is correct.

易错点 / Common trap

50%/4 = 12.5%忽略了增长的复利效应。

50%/4 = 12.5% ignores compounding.

资料提示 / Source note

原答案对选项B的说明写成ln(1.125) - 1 = 0.11778，算式与数值不一致；实际ln(1.125)约为0.11778，不应再减1。正确答案A不受影响。

The source explanation for B writes ln(1.125) - 1 = 0.11778, which is inconsistent. In fact ln(1.125) is approximately 0.11778, without subtracting 1. The correct choice A is unaffected.

1.4 报价利率、有效年利率与连续复利 / Stated rates, effective annual rates, and continuous compounding

先学知识点 / Learn the concept first

名义年利率按季复利时，每季利率等于年报价除以4，一年复利4次。跨年度报价不同，分年计算财富倍数后相乘。

For quarterly compounding, divide the stated annual rate by four and compound four times per year. If the quoted rate changes between years, multiply the separate annual wealth factors.

不同复利频率的产品要比较相同期间的终值或有效年利率。连续复利报价r_c对应财富倍数e^(r_c)。

Compare products with different compounding frequencies using equal-horizon future values or effective annual rates. A continuously compounded annual rate r_c produces the wealth factor e^(r_c).

$$EAR = (1 + r/m)^m - 1; \quad r_c = \ln(1 + EAR)$$

r为名义年利率，m为一年复利次数，EAR为有效年利率。

r is the stated annual rate, m the number of compounding periods per year, and EAR the effective annual rate.

Q54 季度复利与连续复利等价 / Equivalent quarterly and continuous compounding

核心例题 / Core example • 原题 PDF p.125 / 章节 p.11 • 答案 PDF p.54

54. An investor is considering two term deposits with the following characteristics:

	Term Deposit 1	**Term Deposit 2**
Compounding frequency	Quarterly	Continuous
Stated annual rate	4%	?

The stated annual rate for Term Deposit 2 that should make the investor indifferent between the two term deposits is *closest* to:

A. 3.92%.
B. 3.98%.
C. 4.06%.

中文题干与选项 / Chinese translation

某投资者考虑下表两种定期存款。使投资者对两种存款无差异的存款2名义年利率最接近：

项目	存款1	存款2
复利频率	季度	连续
名义年利率	4%	?

A. 3.92%。

B. 3.98%。

C. 4.06%。

答案 / Answer: B

1. 先计算存款1一年后的财富倍数：(1 + 0.04/4)^4 = 1.04060401。

First compute the annual wealth factor for Deposit 1: (1 + 0.04/4)^4 = 1.04060401.

2. 存款2采用连续复利，令e^(r_c) = 1.04060401，所以r_c = 4 ln(1.01) = 3.9801%，选B。

For continuous compounding, set e^(r_c) = 1.04060401. Then r_c = 4 ln(1.01) = 3.9801%, so B is correct.

易错点 / Common trap

4.06%是存款1的有效年利率，不是与之等价的连续复利报价。

4.06% is Deposit 1's effective annual rate, not the equivalent continuously compounded quote.

Q74 变动报价下的季度复利 / Quarterly compounding with changing stated rates

巩固练习 / Reinforcement • 原题 PDF p.130 / 章节 p.16 • 答案 PDF p.74

74. A bank offers a savings account with a stated annual rate of 3% in the first year and 5% in the second year. If returns are compounded quarterly and €90,000 is deposited in the account at the beginning of the first year, the account's value at the end of the second year is *closest* to:
A. 697,200.
B. 697,335.
C. 697,455.

中文题干与选项 / Chinese translation

某银行储蓄账户第一年名义年利率为3%，第二年为5%，按季度复利。若第一年年初存入90,000欧元，第二年年底账户价值最接近：

A. 697,200。

B. 697,335。

C. 697,455。

原答案 C；按印刷数值无匹配项 / Supplied C; no printed option matches

1. 第一年季度利率为0.03/4，第二年季度利率为0.05/4，每年各复利4次。

The quarterly rates are 0.03/4 in Year 1 and 0.05/4 in Year 2, with four compounding periods in each year.

2. FV = 90,000(1 + 0.03/4)^4(1 + 0.05/4)^4 = 97,454.72欧元，约为97,455欧元。

FV = 90,000(1 + 0.03/4)^4(1 + 0.05/4)^4 = €97,454.72, approximately €97,455.

3. 原答案指定C，且解析按约97,455欧元计算；但原题印成697,455，因此三个原样选项均不匹配计算结果。

The source answer identifies C and calculates approximately €97,455, but the printed option is 697,455. None of the choices as printed matches the calculation.

易错点 / Common trap

不能把3%与5%直接相加后乘本金，也不能把名义年利率直接当成每季利率。

Do not simply add 3% and 5% and multiply by principal, or use a stated annual rate as the quarterly rate.

资料提示 / Source note

原题选项存在排印问题：原文A/B/C分别为697,200、697,335、697,455。已保留原文；原答案C应理解为意图对应约97,455欧元，而非认可697,455。

The printed choices contain an apparent typographical error: A/B/C are 697,200, 697,335, and 697,455. They are preserved; source answer C is understood as the intended choice for about €97,455, not as validation of 697,455.

1.5 资金加权与时间加权：先选衡量对象 / Money-weighted versus time-weighted: identify whose performance

先学知识点 / Learn the concept first

资金加权收益率MWR就是使现金流净现值为零的内部收益率IRR。它同时受到现金流金额和发生时间的影响，反映投资者实际投入资金的回报。

The money-weighted return, MWR, is the internal rate of return, IRR, that sets the net present value of cash flows to zero. It depends on cash-flow amounts and timing and measures the return earned on the investor's actual funds.

时间加权收益率TWR按外部资金流分段，再把各段收益率几何链接。客户决定入金出金时，评价基金经理通常使用TWR。

The time-weighted return, TWR, separates performance into subperiods at external cash flows and geometrically links the subperiod returns. Use TWR to evaluate a manager when the client controls contributions and withdrawals.

计算MWR时，支付为负、收到为正；未再投资股息按实际收到的时间列入现金流。中途股票市价若没有发生交易，不是现金流。

For MWR, payments are negative and receipts are positive. Record non-reinvested dividends when received. An interim market price is not a cash flow unless a transaction takes place.

$$0 = CF0 + CF1/(1 + r) + CF2/(1 + r)^2 + ...$$

求使等式成立的r，即资金加权收益率。

Solve for r, the money-weighted return.

Q60 资金加权收益的含义 / Meaning of the money-weighted return

巩固练习 / Reinforcement • 原题 PDF p.127 / 章节 p.13 • 答案 PDF p.60

60. Which of the following statements is *most* accurate? The money-weighted return:
    A. ignores cash withdrawals and additional cash investments.
    B. measures what the investor actually earned on the funds invested.
    C. should be used to compare the performance of different investment managers.

中文题干与选项 / Chinese translation

下列哪项陈述最准确？资金加权收益率：

A. 忽略现金提款及追加投资。

B. 衡量投资者在投入资金上实际赚取的回报。

C. 应当用于比较不同投资经理的表现。

答案 / Answer: B

1. MWR把每次投入和收回资金的金额、时间纳入IRR方程，因此反映投资者实际资金经历的回报。

MWR includes the amount and timing of each investment and withdrawal in the IRR equation, so it captures the return experienced by the investor's actual funds.

2. A与定义相反；经理无法控制客户资金流时，比较经理表现通常用TWR而非MWR。

A contradicts the definition. When a manager cannot control client cash flows, TWR is normally used for manager comparisons rather than MWR.

易错点 / Common trap

“投资者实际赚多少”与“经理投资能力如何”是两个评价目标。

The investor's actual return and the manager's investment performance are different evaluation objectives.

Q61 资金流时机对收益率的影响 / Effect of cash-flow timing on measured returns

巩固练习 / Reinforcement • 原题 PDF p.127 / 章节 p.13 • 答案 PDF p.61

61. In evaluating portfolio performance, the return measure *most* affected by an addition of funds to the portfolio just before a market downturn is the:
    A. time-weighted return.
    B. arithmetic mean return.
    C. money-weighted return.

中文题干与选项 / Chinese translation

评价投资组合表现时，若在市场下跌之前刚好追加资金，以下哪种收益率指标受影响最大?

A. 时间加权收益率。

B. 算术平均收益率。

C. 资金加权收益率。

答案 / Answer: C

1. 追加资金使下跌期间有更多本金承受亏损，改变了投资者的实际资金回报。

The contribution puts more capital at risk during the downturn, changing the return experienced by the investor's funds.

2. MWR对资金金额和时间敏感，故选C。TWR通过分段链接消除外部现金流规模对评价的直接影响。

MWR is sensitive to cash-flow amounts and timing, so C is correct. TWR links subperiod returns to remove the direct effect of external cash-flow size.

易错点 / Common trap

不要把“市场下跌影响所有收益率”与“额外入金的时间最影响哪种口径”混淆。

Do not confuse the effect of a market decline on returns with which measure is most affected by the timing of the additional contribution.

Q91 客户控制资金流时评价经理 / Evaluating a manager when the client controls cash flows

巩固练习 / Reinforcement • 原题 PDF p.133 / 章节 p.19 • 答案 PDF p.91

91. If a client controls the timing of cash flows into and out of a portfolio, which of the following is *most* appropriate when evaluating the performance of the portfolio manager?
    A. Time-weighted rate of return.
    B. Arithmetic mean rate of return.
    C. Money-weighted rate of return.

中文题干与选项 / Chinese translation

若客户决定投资组合资金流入与流出的时间，在评价投资组合经理表现时，最适合使用:

A. 时间加权收益率。

B. 算术平均收益率。

C. 资金加权收益率。

答案 / Answer: A

1. 经理不控制客户入金和提款的时机，因此不应把客户资金流决定造成的影响归入经理表现。

The manager does not control the client's contribution and withdrawal timing, so the effect of those decisions should not determine the manager's measured performance.

2. TWR按外部现金流分段并链接投资回报，故选A。

TWR separates periods at external cash flows and links investment returns, so A is correct.

易错点 / Common trap

MWR可准确反映客户实际体验，却会混合经理表现与客户资金流时机的影响。

MWR describes the client's actual experience but combines manager performance with the effects of client cash-flow timing.

Q26 资金加权收益与股息时间 / Money-weighted return and dividend timing

巩固练习 / Reinforcement • 原题 PDF p.119 / 章节 p.5 • 答案 PDF p.26

26. An investor records the following information and transactions for a stock:

Year	Share at Beginning of Year	Cash Flow at Beginning of Year
1	$100	$100 used to purchase one share
2	$100	$9 received in dividends and not reinvested
3	$100	$100 received from selling one share

The investor's money-weighted rate of return is *closest* to:

A. 4.4%.

B. 4.5%.

C. 4.6%.

中文题干与选项 / Chinese translation

某投资者记录了下表股票资料与交易。其资金加权收益率最接近：

年份	年初股价	年初现金流
1	100美元	支付100美元买入1股
2	100美元	收到9美元股息，不再投资
3	100美元	卖出1股，收到100美元

A. 4.4%。

B. 4.5%。

C. 4.6%。

答案 / Answer: C

1. 年1年初为t=0，年2年初为t=1，年3年初为t=2。现金流分别为-100、+9和+100美元。

The beginning of Year 1 is t = 0, Year 2 is t = 1, and Year 3 is t = 2. The cash flows are -$100, +$9, and +$100.

2. 解100 = 9/(1 + r) + 100/(1 + r)^2，得r = 4.6012%，选C。

Solve 100 = 9/(1 + r) + 100/(1 + r)^2 to obtain r = 4.6012%, so C is correct.

3. 9美元没有再投资，因此必须列在t=1；不能把它移动到t=2并与卖股收入合并。

The $9 dividend was not reinvested, so it belongs at t = 1; do not move it to t = 2 and combine it with the sale proceeds.

易错点 / Common trap

4.5%是两个年度持有期收益率的算术平均；4.4%对应改变股息时间后的计算。

4.5% is the arithmetic average of the two annual holding-period returns; 4.4% arises from changing the dividend timing.

Q59 中途市价不等于现金流 / An interim market price is not a cash flow

巩固练习 / Reinforcement • 原题 PDF p.127 / 章节 p.13 • 答案 PDF p.59

59. An investor gathers the following information about a stock:

Stock price at t=0	$20
Dividend paid at t=1	$3
Stock price at t=1	$12
Dividend paid at t=2	$1
Stock price at t=2	$20

The investor purchased one unit of the stock at t=0 and sold it at t=2. If the dividends were not reinvested, the money-weighted rate of return is *closest* to:

A. 10%.
B. 15%.
C. 20%.

中文题干与选项 / Chinese translation

某投资者获得下表股票资料，于t=0买入1股，并于t=2卖出。若股息不再投资，资金加权收益率最接近：

项目	金额
t=0股价	20美元
t=1股息	3美元
t=1股价	12美元
t=2股息	1美元
t=2股价	20美元

A. 10%。

B. 15%。

C. 20%。

答案 / Answer: A

1. 现金流为CF0 = -20，CF1 = +3，CF2 = +(20 + 1) = +21美元。t=1股价12美元没有伴随交易，不能列作现金流。

The cash flows are CF0 = -20, CF1 = +3, and CF2 = +(20 + 1) = +21 dollars. The $12 price at t = 1 is not a cash flow because no transaction occurs then.

2. 解20 = 3/(1 + r) + 21/(1 + r)^2，得r = 10.2436%，最接近10%，选A。

Solve 20 = 3/(1 + r) + 21/(1 + r)^2 to obtain r = 10.2436%, closest to 10%, so A is correct.

易错点 / Common trap

20%只是全部现金收入相对初始投入的两年累计比例，没有处理资金收到的时间。

20% is only the two-year aggregate cash gain relative to the initial investment; it does not account for receipt timing.

1.6 时间加权收益率的计算顺序 / Calculating a time-weighted return

先学知识点 / Learn the concept first

先逐段算持有期收益，剔除新增投入造成的资产增加，但保留股息等投资收入。再将各段的1+R相乘；若要求多年年化，最后开相应年数的方。

First calculate each subperiod holding-period return, excluding increases caused by additional capital but including investment income such as dividends. Multiply the subperiod wealth factors; annualize over multiple years by taking the appropriate root.

两年收益率的算术平均值一般不等于两年复利年化收益率。外部资金流会影响MWR，但不应直接被当成TWR的收益。

The arithmetic average of two annual returns generally differs from their compound annual rate. External cash flows affect MWR but must not be counted directly as gains in TWR.

TWR_annual = [(1 + R1)(1 + R2)...(1 + RN)]^(1/N) - 1

此式适用于N个年度子期间。

This form applies to N annual subperiods.

Q37 含新增投入的时间加权收益 / Time-weighted return with additional investment

核心例题 / Core example • 原题 PDF p.122 / 章节 p.8 • 答案 PDF p.37

37. An investor summarizes end-of-year cash outlays and proceeds from a two-year investment in a company's shares below.

Year	Outlays	Proceeds
0	$100 to purchase the first share	---
1	$110 to purchase the second share	$10 dividend received from first share (not reinvested)
2	---	$230 received from selling two shares at $115 per share

The annualized time-weighted rate of return of the investment over the two-year period is *closest* to:

A. 9.7%.

B. 12.0%.

C. 12.3%.

中文题干与选项 / Chinese translation

某投资者将一项两年股票投资的年末现金支出和收入汇总如下。该投资两年期间的年化时间加权收益率最接近：

年份	支出	收入
0	支付100美元买入第一股	无
1	支付110美元买入第二股	第一股收到10美元股息，不再投资
2	无	以每股115美元卖出两股，共收到230美元

A. 9.7%。

B. 12.0%。

C. 12.3%。

答案 / Answer: B

1. 第一年：R1 = (110 - 100 + 10)/100 = 20%。股息是收益，另花110美元购买第二股是新增资金。

Year 1: R1 = (110 - 100 + 10)/100 = 20%. The dividend is investment income; the $110 spent on a second share is additional capital.

2. 第二年期初两股价值220美元，期末卖出收到230美元，所以R2 = (230 - 220)/220 = 4.5455%。

The two shares are worth $220 at the beginning of Year 2 and sell for $230 at its end, so R2 = (230 - 220)/220 = 4.5455%.

3. 年化TWR = [(1.20)(1.045455)]^(1/2) - 1 = 12.0065%，选B。

Annualized TWR = [(1.20)(1.045455)]^(1/2) - 1 = 12.0065%, so B is correct.

易错点 / Common trap

9.7%是该组现金流的MWR；12.3%是两年收益的算术平均。TWR要几何链接。

9.7% is the MWR of these cash flows; 12.3% is their arithmetic mean return. TWR requires geometric linking.

单元 02 货币时间价值与年金

Time Value of Money and Annuities

所有现值题都先回答三个问题：现金流在哪些时点发生、每期利率是多少、比较时点在哪里。递延年金的关键是普通年金公式算出的价值位于第一笔付款前一期。

For every present-value problem, identify the cash-flow dates, the interest rate per period, and the valuation date. For a deferred annuity, the ordinary-annuity formula values the stream one period before its first payment.

本单元 6 题 / 6 questions in this unit

单元速记 / Unit essentials

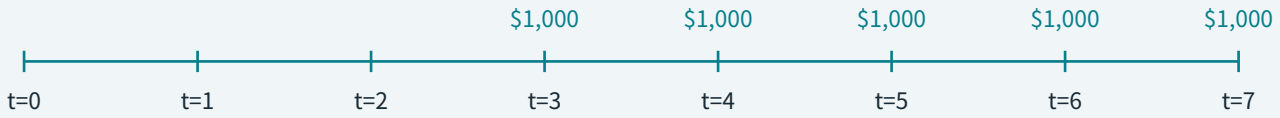
第一笔在一年后：普通年金；第一笔在今天：先付年金；第一笔在第k年：普通年金价值先落在第k-1年。

First payment in one year: ordinary annuity. First payment today: annuity due. First payment at year k: ordinary-annuity value initially lands at year k - 1.

半年度复利要把利率减半、期数加倍；所有方案必须在同一时点比较。

For semiannual compounding, halve the rate and double the period count; compare all alternatives at one date.

原题 Q13：第一笔在 t=3 / First payment at t=3



年金公式先算 t=2，再折现两年。 / Value at t=2 first; then discount two years.

2.1 单笔现金流：利率与期数必须匹配 / Single cash flows: match the rate and period count

先学知识点 / Learn the concept first

若一年复利m次，将名义年利率除以m，并将年数乘以m。两处必须同时调整。

If compounding occurs m times a year, divide the stated annual rate by m and multiply the number of years by m. Both adjustments are necessary.

未来负债折回今天用除法，今天本金滚到未来用乘法。复利频率越高，在相同正名义利率下，给定终值所需现值越低。

Discount a future liability by dividing; compound a current investment by multiplying. At the same positive stated annual rate, more frequent compounding lowers the present value needed for a given future value.

$$PV = FV/(1 + r/m)^{(mN)}$$

N为年数，mN为总复利期数。

N is the number of years and mN is the total number of compounding periods.

Q78 半年复利下的单笔现值 / Single present value with semiannual compounding

巩固练习 / Reinforcement • 原题 PDF p.130 / 章节 p.16 • 答案 PDF p.78

78. A pension fund needs to pay a lump sum $10,000,000 to its participants in 15 years. If the fund is expected to earn 5% per year compounded semi-annually, the amount needed today to meet its liability in 15 years is *closest* to:
    A. $4,767,427.
    B. $4,810,171.
    C. $4,892,771.

中文题干与选项 / Chinese translation

某养老基金需在15年后向参与者一次支付10,000,000美元。若预期收益率为名义年利率5%，每半年复利一次，今天所需金额最接近：

A. 4,767,427美元。

B. 4,810,171美元。

C. 4,892,771美元。

答案 / Answer: A

1. 每半年利率为5%/2 = 2.5%；15年共有15 × 2 = 30个复利期。

The semiannual rate is 5%/2 = 2.5%, and 15 years contains 15 × 2 = 30 compounding periods.

2. PV = 10,000,000/1.025^30 = 4,767,426.85美元，选A。

PV = 10,000,000/1.025^30 = $4,767,426.85, so A is correct.

易错点 / Common trap

只把期数改成30但仍用5%，或只把利率减半但仍用15期，都会错。

Changing the period count to 30 while keeping 5%, or halving the rate while retaining 15 periods, is incorrect.

资料提示 / Source note

原答案的计算器输入行写作I/Y = 0.025。若计算器利率栏按百分数输入，采用每半年一期、N = 30时，应输入2.5，表示2.5%；不要照抄为0.025%。原答案的折现公式和选项A正确。

The source calculator line gives I/Y = 0.025. If the calculator rate field uses percentage input, with each period representing half a year and N = 30, enter 2.5 for 2.5%, not 0.025%. The source discounting formula and choice A are correct.

2.2 普通年金与先付年金 / Ordinary annuities and annuities due

先学知识点 / Learn the concept first

普通年金第一笔在一期后，先付年金第一笔在今天。同样付款次数、金额和正利率下，先付年金现值更高。

An ordinary annuity starts one period from now; an annuity due starts today. With equal payments, payment counts, and a positive interest rate, an annuity due has the higher present value.

比较多个收款方案时，必须把全部现金流换算到同一天。总收款金额最大，不等于现值最大。

Convert every alternative to the same valuation date. The largest undiscounted total payment does not necessarily have the highest present value.

PV_ordinary = PMT[1 - (1 + r)^(-N)]/r

r为每期利率，N为付款次数。

r is the rate per payment period and N is the number of payments.

PV_due = PV_ordinary(1 + r)

先付年金的每笔现金流都比普通年金提前一期。

Each payment of an annuity due occurs one period earlier than its ordinary-annuity counterpart.

Q31 收款方案现值比较 / Comparing the present values of payment alternatives

巩固练习 / Reinforcement · 原题 PDF p.120 / 章节 p.6 · 答案 PDF p.31

31. An investor has three options for receiving payments from an investment:
    - Option 1:a single payment of $136,000 today,
    - Option 2:30 annual payments of $12,000,beginning one year from today,
    - Option 3:20 annual payments of $13,000,beginning today.

    If the annual discount rate is 8%, the option with the highest present value is:

    A. Option 1.
    B. Option 2.
    C. Option 3.

中文题干与选项 / Chinese translation

某投资者可选择三种收款方式：方案1今天一次收取136,000美元；方案2从一年后开始，每年收取12,000美元，共30次；方案3从今天开始，每年收取13,000美元，共20次。若年折现率为8%，哪种方案现值最高？

A. 方案1。

B. 方案2。

C. 方案3。

答案 / Answer: C

1. 方案1的现值直接为136,000美元。方案2为普通年金，PV2 = 12,000[1 - 1.08^(-30)]/0.08 = 135,093.40美元。

Option 1 has a present value of $136,000. Option 2 is an ordinary annuity: PV2 = 12,000[1 - 1.08^(-30)]/0.08 = $135,093.40.

2. 方案3从今天收款，为先付年金：PV3 = 13,000[1 - 1.08^(-20)]/0.08 × 1.08 = 137,846.79美元。

Option 3 starts today and is an annuity due: PV3 = 13,000[1 - 1.08^(-20)]/0.08 × 1.08 = $137,846.79.

3. 比较：137,846.79 > 136,000 > 135,093.40，所以选C。

Comparison: 137,846.79 > 136,000 > 135,093.40, so C is correct.

易错点 / Common trap

方案3若漏乘1.08，会被错误当成普通年金；也不能只比较不折现的总付款额。

Omitting the factor 1.08 incorrectly treats Option 3 as an ordinary annuity; comparing undiscounted totals is also incorrect.

2.3 递延年金：先定位，再折现 / Deferred annuities: locate the value, then discount

先学知识点 / Learn the concept first

第一笔付款在第k年时，普通年金公式得到的是第k-1年的价值。再折现k-1年回今天。

If the first payment occurs at year k, the ordinary-annuity formula produces the value at year k - 1. Discount that value for another k - 1 years to reach today.

也可在第一笔付款当日用先付年金公式，然后折现k年；两条路径必须得到同一个结果。学费等“年初支付”题尤其要先写清每笔付款年份。

Alternatively, use the annuity-due formula at the first payment date and discount for k years. Both methods must agree. For tuition paid at the beginning of each year, list the exact payment years first.

PV0 = PMT[1 - (1 + r)^(-N)]/[r(1 + r)^(k - 1)]

第一笔付款在t=k，共N笔。

The first of N payments occurs at t = k.

Q13 首笔在第3年的递延年金 / Deferred annuity starting in Year 3

核心例题 / Core example • 原题 PDF p.117 / 章节 p.3 • 答案 PDF p.13

13. An investment pays $1,000 annually for five years, with the first payment occurring three years from today. If the discount rate is 6% compounded annually, the present value of the investment today is *closest* to:
    A. $3,537.
    B. $3,749.
    C. $4,212.

中文题干与选项 / Chinese translation

某投资每年支付1,000美元，共五年，第一笔在三年后支付。折现率为按年复利6%，该投资今天的现值最接近：

A. 3,537美元。

B. 3,749美元。

C. 4,212美元。

答案 / Answer: B

1. 付款在t=3、4、5、6、7。普通年金公式先得到第一笔付款前一期，即t=2的价值。

Payments occur at t = 3, 4, 5, 6, and 7. The ordinary-annuity formula first gives the value one period before the first payment, at t = 2.

2. PV2 = 1,000[1 - 1.06^(-5)]/0.06 = 4,212.36美元。

PV2 = 1,000[1 - 1.06^(-5)]/0.06 = $4,212.36.

3. PV0 = 4,212.36/1.06^2 = 3,748.99美元，选B。

PV0 = 4,212.36/1.06^2 = $3,748.99, so B is correct.

易错点 / Common trap

4,212美元在t=2而非今天；再折现3年会多折现一期。

$4,212 is the value at t = 2, not today; discounting it for another three years adds an extra, incorrect period.

Q67 首笔在第5年的递延年金 / Deferred annuity starting in Year 5

巩固练习 / Reinforcement · 原题 PDF p.128 / 章节 p.14 · 答案 PDF p.67

67. An annuity makes seven annual payments of $10,000 each, with the first payment occurring five years from today. If the discount rate is 6% per year, the value of the annuity today is *closest* to:
    A. $41,715.
    B. $44,218.
    C. $55,824.

中文题干与选项 / Chinese translation

某年金每年支付10,000美元，共七次，第一笔在五年后支付。若年折现率为6%，今天的年金价值最接近：

A. 41,715美元。

B. 44,218美元。

C. 55,824美元。

答案 / Answer: B

1. 付款时点为t=5至t=11，普通年金现值应先落在t=4。

The payments occur at t = 5 through t = 11, so the ordinary-annuity value first lands at t = 4.

2. PV4 = 10,000[1 - 1.06^(-7)]/0.06 = 55,823.81美元。

PV4 = 10,000[1 - 1.06^(-7)]/0.06 = $55,823.81.

3. PV0 = 55,823.81/1.06^4 = 44,217.69美元，选B。

PV0 = 55,823.81/1.06^4 = $44,217.69, so B is correct.

易错点 / Common trap

55,824美元是t=4的价值；41,715美元则是多折现一期。

$55,824 is the value at t = 4; $41,715 results from one extra discounting period.

Q66 递延学费的现值 / Present value of deferred tuition payments

巩固练习 / Reinforcement · 原题 PDF p.128 / 章节 p.14 · 答案 PDF p.66

66. An investor needs to make the following payments to cover college tuition fees, starting 10 years from today:

Annual fee (payable at the beginning of each year)	$50,000
Number of years of fee payments	4

If the investor's annual discount rate is 3%, the minimum investment amount required today to fund all four years of college tuition is *closest* to:
    A. $138,294.
    B. $142,442.
    C. $146,716.

中文题干与选项 / Chinese translation

某投资者需从10年后开始支付大学学费，每年年初支付50,000美元，共支付4年。若年折现率为3%，今天至少需要投资的金额最接近：

A. 138,294美元。

B. 142,442美元。

C. 146,716美元。

答案 / Answer: B

1. 实际付款时点是t=10、11、12、13，合计四笔。

The payment dates are t = 10, 11, 12, and 13, for a total of four payments.

2. 普通年金公式先算到t=9：PV9 = 50,000[1 - 1.03^(-4)]/0.03。

Use the ordinary-annuity formula to value them at t = 9: PV9 = 50,000[1 - 1.03^(-4)]/0.03.

3. PV0 = PV9/1.03^9 = 142,442.32美元，选B。等价做法是在t=10用先付年金，再折现10年。

PV0 = PV9/1.03^9 = $142,442.32, so B is correct. Equivalently, value an annuity due at t = 10 and discount it for ten years.

易错点 / Common trap

不要在t=10使用普通年金公式再折现10年，否则会把全部付款推迟一年。

Do not use an ordinary-annuity value at t = 10 and then discount ten years; that shifts every payment one year too late.

2.4 为目标终值反求定期投入 / Solving for contributions to a future target

先学知识点 / Learn the concept first

今天开始投入属于先付年金。若投入期结束后仍要增长到更远的目标日，必须再加入这段复利增长。

Contributions starting today form an annuity due. If the target date is later than the contribution period, include the additional years of compounding.

最不容易错的写法是让每笔投入都滚到同一个目标日，再令这些终值之和等于目标金额。

The safest setup is to compound each contribution to the same target date and equate their combined future values to the target.

> Target FV at T = PMT × sum[(1 + r)^(T - t)]
>
> 对每个实际投入时点t求和；不要把最后一笔投入时点与投入次数混淆。
>
> Sum over the actual contribution dates t; do not confuse the date of the last contribution with the number of contributions.

Q63 先付投入与远期目标 / Contributions beginning today for a later target

核心例题 / Core example • 原题 PDF p.127 / 章节 p.13 • 答案 PDF p.63

63. An investment requires 10 equal annual payments, starting today, and will pay out a lump sum of $500,000 15 years from today. If the interest rate is 4% per year compounded annually, the required annual payment is *closest* to:
    A. $32,913.
    B. $34,230.
    C. $40,044.

中文题干与选项 / Chinese translation

某投资要求投资者每年等额支付，共10次，从今天开始；该投资将于15年后向投资者一次支付500,000美元。若利率为按年复利4%，所需每年支付金额最接近：

A. 32,913美元。

B. 34,230美元。

C. 40,044美元。

答案 / Answer: A

1. 10次投入发生在t=0至t=9；目标终值位于t=15。最后一笔投入仍可增长6年。

The ten contributions occur at t = 0 through t = 9, and the target is at t = 15. The last contribution still compounds for six years.

2. 先将投入累积至t=10：FV10 = PMT[(1.04^10 - 1)/0.04] × 1.04，再乘1.04^5累积至t=15。

First accumulate the contributions to t = 10: FV10 = PMT[(1.04^10 - 1)/0.04] × 1.04. Then multiply by 1.04^5 to reach t = 15.

3. PMT = 500,000 / {[(1.04^10 - 1)/0.04] × 1.04^6} = 32,913.02美元，选A。

PMT = 500,000 / {[(1.04^10 - 1)/0.04] × 1.04^6} = $32,913.02, so A is correct.

易错点 / Common trap

今天开始意味着先付年金；第10次投入在t=9，不在t=10。

Starting today means an annuity due; the tenth contribution occurs at t = 9, not t = 10.

单元 03 描述统计：平均数、位置、离散与相关

Descriptive Statistics: Center, Position, Dispersion and Correlation

先判断题目要描述什么：中心水平、相对位置、波动、尾部形状还是两变量共同变化。识别关键词后再选统计量，比盲目代公式更快。

First identify what the question describes: central level, relative position, dispersion, tail shape, or co-movement. Match the keyword to the statistic before substituting numbers.

本单元 19 题 / 19 questions in this unit

单元速记 / Unit essentials

先问“求位置还是求数值”：第几个观测和百分位数值不是同一回事。

Distinguish an observation location from the numerical value of a percentile.

先问“样本还是概率分布”：样本方差用 n-1；已知概率分布直接按概率加权。

Distinguish a sample from a probability distribution: sample variance uses n-1; known distributions use probability weights.

风险指标分子分母要读清：CV 是风险/平均收益，Sharpe 是超额收益/风险。

Read ratio direction carefully: CV is risk/mean return; Sharpe is excess return/risk.

3.1 平均数怎么选 / Choosing a mean

先学知识点 / Learn the concept first

算术平均数给每个观测值相同权重；几何平均数用于连乘增长因子；调和平均数适合每期投入相同金额时的平均买入单价。

The arithmetic mean weights observations equally; the geometric mean combines multiplicative growth factors; the harmonic mean gives average purchase cost when equal amounts of money are invested.

对严格为正且不完全相同的数值，调和平均数 < 几何平均数 < 算术平均数。若所有数值相同，三者相等。出现零或负数时，先检查公式是否有定义。

For strictly positive observations that are not all equal, harmonic mean < geometric mean < arithmetic mean. All three are equal when all values are equal. With zeros or negative values, check the domain before applying a formula.

截尾平均数删除两端极值；缩尾平均数保留观测个数，但把极端值替换为指定分位点附近的值。

A trimmed mean removes tail observations. A winsorized mean keeps the sample size but replaces extreme observations with values at specified cutoffs.

AM = sum(x_i)/n; GM = (product(x_i))^(1/n); HM = n/sum(1/x_i)

AM、GM、HM 分别表示算术、几何与调和平均数。

AM, GM and HM denote the arithmetic, geometric and harmonic means.

Average cost = Total money invested / Total shares purchased

等额投入时，不是对每期股价直接取算术平均。

With equal cash investments, do not simply average the periodic share prices.

Q39 三种平均数的大小关系 / Ordering three means

巩固练习 / Reinforcement • 原题 PDF p.122 / 章节 p.8 • 答案 PDF p.39

39. For a given dataset with different non-negative observations, which of the following will have the largest value?
    A. Harmonic mean.
    B. Arithmetic mean.
    C. Geometric mean.

中文题干与选项 / Chinese translation

对于由不同的非负观测值组成的数据集，下列哪个平均数最大?

A. 调和平均数。

B. 算术平均数。

C. 几何平均数。

答案 / Answer: B

1. 按本题意图，对严格为正且不完全相同的观测：HM < GM < AM，故选算术平均数。

Under the intended positive-data setting, observations that are not all equal satisfy HM < GM < AM, so choose the arithmetic mean.

2. 不能凭某个平均数名字判断大小；应先识别适用域，再记住 HM、GM、AM 的顺序。

Check the domain first, then apply the HM, GM, AM ordering.

易错点 / Common trap

包含零时，标准的 HM=n/sum(1/x_i) 不能直接计算。

With a zero observation, the usual HM=n/sum(1/x_i) formula cannot be applied directly.

资料提示 / Source note

原题写“非负”而非“严格为正”，调和平均数的通常公式需要所有观测大于零。答案 B 为原解析的预期答案；此处补充适用条件，不改原题。

The stem says non-negative rather than strictly positive, although the usual harmonic-mean formula requires positive observations. B is the intended supplied answer; this note states the domain without changing the question.

Q55 等额投入的平均成本 / Average cost under equal cash investments

核心例题 / Core example • 原题 PDF p.126 / 章节 p.12 • 答案 PDF p.55

55. An investor invests a fixed amount of money into a fund each year for three years as follows:

Year	Term Deposit 1
1	€ 14.00
2	€ 12.00
3	€ 17.00

The investor's average cost per share is *closest* to:

A. € 14.05.

B. € 14.33.

C. € 14.63.

中文题干与选项 / Chinese translation

投资者连续三年每年投入固定金额购买一只基金，数值如下表。投资者的每股平均成本最接近多少?

年份	定期存款 1（原表标题）
1	€14.00
2	€12.00
3	€17.00

A. €14.05。

B. €14.33。

C. €14.63。

答案 / Answer: A

1. 设每年固定投入金额为 M，三年买入份额分别是 M/14、M/12、M/17；总投入为 3M。

Let the fixed annual investment be M. Units purchased are M/14, M/12 and M/17, while total cash invested is 3M.

2. 平均成本=3M/[M/14+M/12+M/17]=3/(1/14+1/12+1/17)=€14.0459，选 €14.05。

Average cost=3M/[M/14+M/12+M/17]=3/(1/14+1/12+1/17)=€14.0459, closest to €14.05.

3. 价格低时等额资金买到更多份额，因此不能给三个价格相同的份额权重。

Equal cash buys more units at lower prices, so the three prices should not receive equal unit weights.

易错点 / Common trap

等额现金用调和平均；等份额购买才对价格取算术平均。

Equal cash purchases use the harmonic mean; equal numbers of units use the arithmetic mean of prices.

资料提示 / Source note

原表数值列标题误写为“Term Deposit 1（定期存款 1）”；题干与答案均按基金每股买入价处理。保留原数值，不补造投入金额。

The original numeric-column header says “Term Deposit 1”, but the stem and solution treat the amounts as fund purchase prices per unit. Values are preserved and no investment amount is invented.

Q71 截尾平均数 / Trimmed mean

巩固练习 / Reinforcement • 原题 PDF p.129 / 章节 p.15 • 答案 PDF p.71

71. An analyst discards the lowest 2.5% and the highest 2.5% of values in a sample, and computes the mean of the remaining 95% of values. The resulting mean is best described as a:
    A. trimmed mean.
    B. harmonic mean.
    C. winsorized mean.

中文题干与选项 / Chinese translation

分析师删除样本中最低的 2.5% 和最高的 2.5% 数值，再计算剩余 95% 数值的平均数。所得平均数最恰当地称为：

A. 截尾平均数。

B. 调和平均数。

C. 缩尾平均数。

答案 / Answer: A

1. 题目明确删除两端观测，因此是截尾平均数 trimmed mean。

The tail observations are deleted, so this is a trimmed mean.

2. 缩尾平均数 winsorized mean 会替换极端值而保留观测个数；调和平均数则使用倒数。

A winsorized mean replaces extreme values while retaining observations; a harmonic mean uses reciprocals.

易错点 / Common trap

删除是 trimmed；替换是 winsorized。

Deletion means trimmed; replacement means winsorized.

3.2 分位数：先排序，再找位置 / Quantiles: sort first, then locate

先学知识点 / Learn the concept first

第二四分位数 Q2 就是中位数，也是第 50 百分位数。奇数个观测取正中间一个；偶数个观测取中间两个数的平均。

The second quartile Q2 is the median, or 50th percentile. For an odd number of observations, take the middle value; for an even number, average the two middle values.

本题库采用百分位位置 L=(n+1)p，其中 p 用小数表示。位置不是数据值；位置非整数时在相邻观测之间插值。不同软件可能使用不同分位数约定。

This question set uses percentile location L=(n+1)p, with p expressed as a decimal. A location is not a data value; interpolate between adjacent observations when needed. Software packages may use different quantile conventions.

十分位组各占 10%，五分位组各占 20%，四分位组各占 25%。本资料把分界值归入较低一组，因此恰在第 25 百分位仍属于第一四分位组。

Decile groups span 10%, quintile groups 20%, and quartile groups 25%. The supplied solution assigns a boundary to the lower group, so exactly the 25th percentile remains in the first quartile group.

> $L_p = (n+1)p$; IQR = Q3 - Q1
>
> 四分位距只衡量中间 50% 的跨度。
>
> The interquartile range measures the span of the middle 50%.

Q47 排序后的中位数 / Median after sorting

巩固练习 / Reinforcement • 原题 PDF p.124 / 章节 p.10 • 答案 PDF p.47

47. An analyst gathers the following returns for seven funds:

12%	7%	5%	4%	8%	3%	3%

The second quartile return is:

A. 4%.
B. 5%.
C. 6%.

中文题干与选项 / Chinese translation

分析师取得七只基金收益率：12%、7%、5%、4%、8%、3%、3%。第二四分位数收益率是多少?

A. 4%。

B. 5%。

C. 6%。

答案 / Answer: B

1. 先排序为 3%、3%、4%、5%、7%、8%、12%。

Sort the returns: 3%, 3%, 4%, 5%, 7%, 8%, 12%.

2. Q2 为中位数，位置=(7+1)/2=4，第 4 个收益率是 5%。

Q2 is the median at position (7+1)/2=4, whose return is 5%.

3. 4 是位置而不是收益率；6% 是算术平均数。

4 is the location, not the return; 6% is the arithmetic mean.

易错点 / Common trap

不要直接取原始顺序中的中间一个数。

Do not take the middle entry before sorting.

Q43 偶数样本的中位数 / Median of an even-sized sample

巩固练习 / Reinforcement • 原题 PDF p.123 / 章节 p.9 • 答案 PDF p.43

43. An analyst observes the following EPS for four companies: -£0.50, £0.50, £2.50, and £5.50. The 50th percentile of the EPS values is *closest* to:

A. £1.50.
B. £2.00.
C. £2.50.

中文题干与选项 / Chinese translation

分析师观察到四家公司的每股收益分别为 -£0.50、£0.50、£2.50 和 £5.50。其第 50 百分位数最接近多少?

A. £1.50。

B. £2.00。

C. £2.50。

答案 / Answer: A

1. 四个值已按升序排列。第 50 百分位数是中位数，取第二和第三个值的平均。

The four observations are already sorted. The 50th percentile is the median, found by averaging the second and third values.

2. 中位数=(£0.50+£2.50)/2=£1.50。£2.00 是四个数的算术平均，不是中位数。

Median=(£0.50+£2.50)/2=£1.50. £2.00 is the arithmetic mean, not the median.

易错点 / Common trap

百分位位置 2.5 表示在第二与第三个值之间，并不表示 EPS=£2.50。

A percentile location of 2.5 lies between observations two and three; it does not mean EPS equals £2.50.

Q38 分位组与边界 / Quantile groups and boundaries

巩固练习 / Reinforcement • 原题 PDF p.122 / 章节 p.8 • 答案 PDF p.38

38. Ranked in ascending order, the 19h observation in a sample of 75 is in the second:
    A. decile.
    B. quintile.
    C. quartile.

中文题干与选项 / Chinese translation

将 75 个观测按升序排列后，第 19 个观测位于第二个哪一种分位组?

A. 十分位组。

B. 五分位组。

C. 四分位组。

答案 / Answer: B

1. 按原解析的位置公式 L=(n+1)p，p=19/(75+1)=25%。

Using the supplied location convention L=(n+1)p gives p=19/(75+1)=25%.

2. 第二五分位组涵盖大于 20% 且不超过 40% 的位置，所以第 25 百分位落在其中。

The second quintile covers positions above 20% and at or below 40%, so the 25th percentile lies in it.

3. 本解析把恰好第 25 百分位归入第一四分位组；第二四分位组从高于 25% 开始。

The supplied solution puts the exact 25th-percentile boundary in the first quartile group; the second quartile begins above 25%.

易错点 / Common trap

不是“第 19 个”就等于“第 19 百分位”；位置必须按样本量换算。

The 19th observation is not automatically the 19th percentile; convert the location using sample size.

资料提示 / Source note

原题印作“19h”，按答案中的“19th”理解为第 19 个；分位边界遵循本资料的约定。

The original prints “19h”; the answer identifies it as the 19th observation. Quantile boundaries follow the supplied convention.

Q35 四分位距 / Interquartile range

巩固练习 / Reinforcement • 原题 PDF p.121 / 章节 p.7 • 答案 PDF p.35

35. An analyst calculates the following statistics for a sample with 100 observations:

	Value
First quartile	11
Second quartile	62
Third quartile	93
Fourth quartile	359

The interquartile range of the sample is equal to:

A. 31.
B. 82.
C. 348.

中文题干与选项 / Chinese translation

分析师为含 100 个观测的样本计算出下表统计量。该样本的四分位距等于多少?

统计量	数值
第一四分位数	11
第二四分位数	62
第三四分位数	93
第四四分位数（原文）	359

A. 31。

B. 82。

C. 348。

答案 / Answer: B

1. 四分位距 IQR=Q3-Q1=93-11=82。

The interquartile range is IQR=Q3-Q1=93-11=82.

2. 31 是 Q3-Q2；348 是表中第四与第一四分位数值之差，均不是 IQR。

31 is Q3-Q2; 348 is the difference between the fourth and first quartile values shown. Neither is the IQR.

易错点 / Common trap

中间 50% 的宽度只用 Q1 与 Q3。

The width of the middle 50% uses Q1 and Q3 only.

3.3 离散程度与下行风险 / Dispersion and downside risk

先学知识点 / Learn the concept first

平均绝对离差 MAD 对偏离均值的绝对值取平均；样本标准差把偏差平方后除以 n-1，再开平方。标准差与原数据具有相同单位。

Mean absolute deviation averages absolute deviations from the mean. Sample standard deviation squares deviations, divides by n-1, and takes the square root. Standard deviation has the same units as the data.

本题库的样本目标下行偏差只累计低于目标 B 的平方缺口，但分母仍是全部样本个数减一；高于目标的观测对分子贡献为零。

The sample target downside deviation used here includes only squared shortfalls below target B, but its denominator remains total sample size minus one. Observations above the target contribute zero to the numerator.

变异系数 CV 表示每单位平均收益承担多少标准差。它与夏普比率方向不同；夏普比率衡量每单位风险得到多少超额收益。均值为零或负数时，不宜机械比较 CV。

The coefficient of variation measures standard deviation per unit of mean return. The Sharpe ratio instead measures excess return per unit of risk. Avoid mechanical CV comparisons when the mean is zero or negative.

MAD = sum(abs(x_i - mean))/n; s = sqrt(sum((x_i - mean)^2)/(n-1))

MAD 用 n；本资料的样本标准差用 n-1。

MAD uses n; the sample standard deviation here uses n-1.

Target downside deviation = sqrt(sum(min(0, R_i - B)^2)/(n-1))

n 为全部观测个数，不是低于目标的观测个数。

n is the total number of observations, not the number below the target.

CV = s/mean; Sharpe = (mean return - risk-free rate)/s

风险除以平均收益是 CV；超额收益除以风险是 Sharpe。

Risk divided by mean return is CV; excess return divided by risk is the Sharpe ratio.

Q58 平均绝对离差与样本标准差 / Mean absolute deviation and sample standard deviation

巩固练习 / Reinforcement · 原题 PDF p.126 / 章节 p.12 · 答案 PDF p.58

58. An analyst gathers the following sample returns for a security:

Return
-2%
-1%
1%
2%

The mean absolute deviation of the sample returns is:

A. less than the sample standard deviation.

B. equal to the sample standard deviation.

C. greater than the sample standard deviation.

中文题干与选项 / Chinese translation

分析师取得某证券的样本收益率：-2%、-1%、1%、2%。这些样本收益率的平均绝对离差：

A. 小于样本标准差。

B. 等于样本标准差。

C. 大于样本标准差。

答案 / Answer: A

1. 平均收益为 0%。MAD=(2+1+1+2)/4=1.5%。

The mean return is 0%. MAD=(2+1+1+2)/4=1.5%.

2. 样本标准差=sqrt((4+1+1+4)/(4-1))=sqrt(10/3)=1.8257%。所以 MAD 更小。

Sample standard deviation=sqrt((4+1+1+4)/(4-1))=sqrt(10/3)=1.8257%, which exceeds MAD.

易错点 / Common trap

平方后求平均再开方，与先取绝对值再求平均不是同一操作。

A root mean square is not the same calculation as a mean absolute deviation.

Q45 样本目标下行偏差 / Sample target downside deviation

核心例题 / Core example · 原题 PDF p.123 / 章节 p.9 · 答案 PDF p.45

45. An analyst gathers the following information about a portfolio's returns:

Year	Return (%)
1	6%
2	7%
3	3%
4	2%
5	4%

If the target return is 5%, the target downside deviation is *closest* to:

A. 1.7%.

B. 1.9%.

C. 2.2%.

中文题干与选项 / Chinese translation

分析师取得下表组合收益率。若目标收益为 5%，目标下行偏差最接近多少?

年份	收益率
1	6%
2	7%
3	3%
4	2%
5	4%

A. 1.7%。

B. 1.9%。

C. 2.2%。

答案 / Answer: B

1. 以百分数数值计算，低于 5% 的缺口为 -2、-3、-1；高于目标的 6% 和 7% 对分子贡献为零。

Using percentage-point inputs, shortfalls below 5% are -2, -3 and -1. The 6% and 7% returns contribute zero to the numerator.

2. 平方缺口和=4+9+1=14；全部样本数 n=5，所以按原解析，下行偏差=sqrt(14/(5-1))=1.8708%，选 1.9%。

The squared-shortfall sum is 4+9+1=14. Total sample size is n=5, so the supplied sample convention gives sqrt(14/(5-1))=1.8708%, closest to 1.9%.

3. 用 n=5 作分母会得到约 1.7%；把高于目标的偏差也算进去会得到约 2.2%，对应两个干扰项。

Dividing by n=5 gives about 1.7%; including above-target deviations gives about 2.2%, explaining the distractors.

易错点 / Common trap

分母是全部观测数减一，不是低于目标的观测数。

The denominator is total sample size minus one, not the count of below-target observations.

资料提示 / Source note

本讲义按用户提供答案采用样本口径 n-1；实际应用中也有以 n 为分母的下行偏差定义，须先确认约定。

This guide follows the supplied answer's sample convention n-1. Some applications define downside deviation with n, so identify the convention first.

Q70 目标等于均值时的半标准差 / Semideviation when the target equals the mean

巩固练习 / Reinforcement • 原题 PDF p.129 / 章节 p.15 • 答案 PDF p.70

70. A portfolio has a mean return of 1.0% and a standard deviation of returns of 2.7%. If the specified minimum target return is 1.0%, the sample target semideviation is:
    A. less than 2.7%.
    B. equal to 2.7%.
    C. greater than 2.7%.

中文题干与选项 / Chinese translation

某组合平均收益率为 1.0%，收益率标准差为 2.7%。若最低目标收益为 1.0%，样本目标半标准差：

A. 小于 2.7%。

B. 等于 2.7%。

C. 大于 2.7%。

答案 / Answer: A

1. 目标恰好等于均值，半标准差只保留低于均值的平方偏差，而标准差包括全部平方偏差。

The target equals the mean. Semideviation includes only squared deviations below the mean, whereas standard deviation includes all squared deviations.

2. 采用相同的样本分母 n-1 时，半标准差的分子更小，因此结果小于 2.7%。

With the same sample denominator n-1, the semideviation numerator is smaller, giving a value below 2.7%.

易错点 / Common trap

不能仅凭标准差求出半标准差的精确值，也不能直接除以 2。

The exact semideviation cannot be inferred from standard deviation alone; do not simply divide by two.

Q46 变异系数与夏普比率 / Coefficient of variation versus Sharpe ratio

巩固练习 / Reinforcement • 原题 PDF p.123 / 章节 p.9 • 答案 PDF p.46

46. Which of the following measures best quantifies the amount of risk per unit of mean return?
    A. Sharpe ratio
    B. Standard deviation
    C. Coefficient of variation

中文题干与选项 / Chinese translation

下列哪一项最能量化每单位平均收益所承担的风险?

A. 夏普比率。

B. 标准差。

C. 变异系数。

答案 / Answer: C

1. 变异系数 CV=标准差/平均收益，直接回答“每单位平均收益承担多少风险”。

CV=standard deviation/mean return directly measures risk per unit of mean return.

2. 标准差仅衡量风险；夏普比率=(平均收益-无风险利率)/标准差，衡量每单位风险的超额收益。

Standard deviation measures risk alone. Sharpe=(mean return-risk-free rate)/standard deviation measures excess return per unit of risk.

易错点 / Common trap

先读清 per unit 后面是什么；分母就是“每单位”的对象。

Read what follows per unit: that quantity belongs in the denominator.

Q81 变异系数的方向 / Direction of the coefficient of variation

巩固练习 / Reinforcement • 原题 PDF p.131 / 章节 p.17 • 答案 PDF p.81

81. For a set of return observations, the coefficient of variation is best described as a measure of.
    A. risk per unit of mean return.
    B. mean excess return earned per unit of risk.
    C. average absolute deviation around the mean return.

中文题干与选项 / Chinese translation

对于一组收益率观测，变异系数最恰当地衡量：

A. 每单位平均收益所承担的风险。

B. 每单位风险获得的平均超额收益。

C. 围绕均值的平均绝对偏差。

答案 / Answer: A

1. CV=标准差/平均收益，即每单位平均收益的风险。

CV=standard deviation/mean return, or risk per unit of mean return.

2. B 描述夏普比率；C 描述平均绝对离差。

B describes the Sharpe ratio, and C describes mean absolute deviation.

易错点 / Common trap

与第 46 题是同一考点，记住比率方向即可一起掌握。

This tests the same concept as Q46; mastering the ratio direction handles both.

3.4 偏度：看长尾，不看峰在哪边 / Skewness: identify the long tail

先学知识点 / Learn the concept first

负偏是左尾较长，常见投资表现为较多小盈利、少数巨大亏损；正偏是右尾较长，典型地有少数巨大盈利。

Negative skew means a longer left tail, often associated with many small gains and a few large losses. Positive skew means a longer right tail, typically with a few large gains.

按本题采用的典型单峰形态，负偏时均值 < 中位数 < 众数；正偏时众数 < 中位数 < 均值。极端值把均值拉向长尾。

For the conventional unimodal shapes used in these questions, negative skew gives mean < median < mode, and positive skew gives mode < median < mean. Extreme observations pull the mean toward the long tail.

偏度讨论左右不对称；峰度讨论尾部厚度等形态特征。仅凭左尾较长，不能判定是尖峰厚尾分布。

Skewness concerns asymmetry; kurtosis concerns features such as tail heaviness. A longer left tail alone does not establish leptokurtosis.

Q18 负偏分布的中心位置 / Center of a negatively skewed distribution

巩固练习 / Reinforcement • 原题 PDF p.118 / 章节 p.4 • 答案 PDF p.18

18. If a unimodal return distribution is negatively skewed, which of the following *most likely* has the highest value?
    A. Mean
    B. Mode
    C. Median

中文题干与选项 / Chinese translation

如果一个单峰收益率分布呈负偏，下列哪个统计量的数值最可能最大?

A. 均值

B. 众数

C. 中位数

答案 / Answer: B

1. 本题采用典型负偏单峰形态：均值 < 中位数 < 众数，因此最大的是众数。

The conventional negatively skewed unimodal shape used here has mean < median < mode, so the mode is highest.

2. 少数很低的收益把均值拉向左尾，均值不会是最大值。

A few very low returns pull the mean toward the left tail, so the mean is not the highest measure.

易错点 / Common trap

负偏的“负”指左尾；不要理解成所有收益均为负。

Negative skew refers to the left tail, not to all returns being negative.

Q21 正偏分布的中心位置 / Center of a positively skewed distribution

巩固练习 / Reinforcement • 原题 PDF p.118 / 章节 p.4 • 答案 PDF p.21

21. For a continuous positively skewed unimodal distribution:
    A. both the mode and the median are less than the mean.
    B. both the mode and the median are greater than the mean.
    C. the mode is less than the mean and the median is greater than the mean.

中文题干与选项 / Chinese translation

对于连续、正偏的单峰分布：

A. 众数和中位数都小于均值。

B. 众数和中位数都大于均值。

C. 众数小于均值，而中位数大于均值。

答案 / Answer: A

1. 本题采用典型正偏单峰形态：众数 < 中位数 < 均值。

The conventional positively skewed unimodal shape used here has mode < median < mean.

2. 右尾的极端大值向上拉高均值，所以众数与中位数都低于均值。

Extreme values in the right tail pull the mean upward, leaving the mode and median below it.

易错点 / Common trap

把均值想成最容易被尾部极端值拉动的位置。

Think of the mean as the measure most easily pulled by extreme tail values.

Q85 负偏收益的经济含义 / Economic meaning of negative skew

巩固练习 / Reinforcement • 原题 PDF p.132 / 章节 p.18 • 答案 PDF p.85

85. A return distribution with negative skew and a mean of zero *most likely* has:
    A. frequent small gains and a few extreme losses.
    B. frequent small losses and a few extreme gains.
    C. frequent extreme losses and a few small gains.

中文题干与选项 / Chinese translation

一个均值为零且负偏的收益率分布最可能表现为：

A. 频繁的小额盈利与少量极端亏损。

B. 频繁的小额亏损与少量极端盈利。

C. 频繁的极端亏损与少量小额盈利。

答案 / Answer: A

1. 负偏意味着左侧亏损尾部更长，少数极大亏损可抵消较多小盈利，使均值为零。

Negative skew means a longer loss tail. A few large losses can offset many small gains, leaving a zero mean.

2. B 是典型正偏描述；C 把极端亏损说成频繁发生，与本题的负偏形态不符。

B is the conventional positive-skew description. C makes extreme losses frequent, unlike the negative-skew pattern intended here.

易错点 / Common trap

均值为零不表示盈利和亏损的次数各占一半。

A zero mean does not imply equal numbers of gains and losses.

Q89 由长尾识别偏度 / Identifying skew from the longer tail

巩固练习 / Reinforcement • 原题 PDF p.132 / 章节 p.18 • 答案 PDF p.89

89. A graphical depiction of a continuous distribution shows the left tail to be longer than the right tail. The distribution is *best* described as having:
    A. negative skewness.
    B. leptokurtosis.
    C. Positive skewness.

中文题干与选项 / Chinese translation

一个连续分布的图形显示左尾比右尾更长。该分布最恰当地描述为：

A. 负偏。

B. 尖峰厚尾。

C. 正偏。

答案 / Answer: A

1. 左尾较长对应负偏，也称左偏。

A longer left tail indicates negative, or left, skewness.

2. 右尾较长才是正偏；仅给出左右尾部长度关系，不足以判定峰度。

A longer right tail would indicate positive skew; relative left/right tail length alone does not determine kurtosis.

易错点 / Common trap

偏度方向由长尾决定，不由峰的位置决定。

The long tail determines skew direction, not the peak location.

3.5 相关系数：标准化的线性关系 / Correlation: standardized linear association

先学知识点 / Learn the concept first

皮尔逊相关系数位于 -1 到 +1，描述线性关系的方向和强度。零相关不代表没有非线性关系，也不自动代表独立。

Pearson correlation ranges from -1 to +1 and measures the direction and strength of linear association. Zero correlation does not rule out a nonlinear relationship or automatically imply independence.

异常值可能显著改变相关系数。协方差与相关系数也不能混用：相关系数还要除以两个标准差。

Outliers may substantially change correlation. Covariance and correlation are not interchangeable: correlation also divides by the two standard deviations.

> rho(X,Y) = Cov(X,Y)/(sigma_X * sigma_Y)
>
> 在正协方差不变时，标准差增大将使相关系数降低。
>
> With a fixed positive covariance, higher standard deviations reduce correlation.

Q48 相关系数的含义 / Meaning of correlation

巩固练习 / Reinforcement • 原题 PDF p.124 / 章节 p.10 • 答案 PDF p.48

48. The correlation coefficient:
    A. ranges from 0 to 1.
    B. is not affected by outliers.
    C. indicates the strength of the linear relationship between two random variables.

中文题干与选项 / Chinese translation

相关系数：

A. 介于 0 和 1 之间。

B. 不受异常值影响。

C. 表示两个随机变量之间线性关系的强度。

答案 / Answer: C

1. 相关系数表示线性关系的方向与强度，取值范围为 -1 到 +1。

Correlation measures the direction and strength of a linear relationship and ranges from -1 to +1.

2. 因此 A 的范围不完整；B 错在异常值可能明显改变相关系数。

A gives an incomplete range; B is wrong because outliers can materially affect correlation.

易错点 / Common trap

相关系数不是只取非负值的 R^2。

Correlation is not the non-negative R-squared statistic.

Q50 相关系数只衡量线性联系 / Correlation measures linear association

巩固练习 / Reinforcement • 原题 PDF p.124 / 章节 p.10 • 答案 PDF p.50

50. The correlation between two variables measures:
    A. only their linear relationship.
    B. only their non-linear relationship.
    C. both their linear and non-linear relationships.

中文题干与选项 / Chinese translation

两个变量之间的相关系数衡量：

A. 仅它们的线性关系。

B. 仅它们的非线性关系。

C. 它们的线性与非线性关系。

答案 / Answer: A

1. 这里指皮尔逊相关系数，它度量线性联系。

This question refers to Pearson correlation, which measures linear association.

2. 存在非线性关系时，相关系数可能很小，因此不能把它当作所有关系的完整度量。

With a nonlinear relationship, correlation may be small, so it is not a complete measure of every kind of dependence.

易错点 / Common trap

零相关不等于完全没有关系。

Zero correlation does not mean no relationship of any kind.

Q49 协方差固定时的相关系数 / Correlation with fixed covariance

巩固练习 / Reinforcement • 原题 PDF p.124 / 章节 p.10 • 答案 PDF p.49

49. If the covariance between two positively correlated random variables remains the same but the variance of both variables increases, the correlation between the two variables.
    A. decreases.
    B. stays the same.
    C. increases.

中文题干与选项 / Chinese translation

若两个正相关随机变量之间的协方差保持不变，而两变量的方差均增加，则相关系数会如何变化？

A. 下降。

B. 保持不变。

C. 上升。

答案 / Answer: A

1. rho=Cov(X,Y)/(sigma_X*sigma_Y)。方差增加意味着标准差增加。

rho=Cov(X,Y)/(sigma_X*sigma_Y). Higher variances imply higher standard deviations.

2. 分子为固定的正值，分母增大，因此相关系数下降。

The numerator stays fixed and positive while the denominator rises, so correlation decreases.

易错点 / Common trap

题目明确固定协方差，不能自行假设它跟着波动一起上升。

The stem fixes covariance; do not assume it rises with volatility.

单元 04 概率、条件期望与联合分布

Probability, Conditional Expectation and Joint Distributions

这组题的核心是认清权重：等可能时用等权，不等可能时用概率权重；给定某情景时，只在该情景内部计算。

The key is choosing the right weights: equal weights for equally likely outcomes, probability weights otherwise, and within-scenario probabilities when conditioning on a scenario.

本单元 5 题 / 5 questions in this unit

单元速记 / Unit essentials

条件期望不乘外层情景概率；无条件期望才把各情景合并。

Do not multiply by the outer scenario probability for a conditional expectation; do so when combining scenarios.

方差题停在平方单位；标准差题最后开平方。

For variance, stop at squared units; for standard deviation, take the square root.

协方差是“偏差的乘积”，不是“两个收益率相减”。

Covariance uses products of deviations, not differences between two returns.

4.1 概率树与条件期望 / Probability trees and conditional expectation

先学知识点 / Learn the concept first

概率树把“先发生什么情景、该情景下再出现什么结果”逐层展开。同一路径的联合概率等于沿路径的概率相乘；不同互斥路径的概率相加。

A probability tree lays out scenarios followed by outcomes within each scenario. Multiply probabilities along a path to obtain its joint probability; add probabilities across mutually exclusive paths.

看到 given、under 或 conditional on，先固定该情景。计算条件期望时，不再乘该情景的发生概率；计算无条件期望时才跨情景加权。

Words such as given, under, or conditional on fix the scenario. Do not multiply a conditional expectation by the scenario probability unless building an unconditional expectation.

$$E(X|S) = \text{sum}[P(X=x_i|S) * x_i]$$

竖线表示“已知情景 S”。

The vertical bar means given scenario S.

$$E(X) = \text{sum}[P(S_j) * E(X|S_j)]$$

全期望公式先算每个情景内的期望，再按情景概率加权。

The total-expectation rule first computes each conditional mean, then weights those means by scenario probabilities.

Q28 概率树的用途 / Use of probability trees

巩固练习 / Reinforcement · 原题 PDF p.120 / 章节 p.6 · 答案 PDF p.28

28. An analyst estimates the probabilities of three possible economic scenarios and the probabilities of a stock having a positive or a negative return in each scenario. These scenarios are best represented by a:
    A. tree-map.
    B. tree diagram.
    C. probability density function.

中文题干与选项 / Chinese translation

分析师估计三种可能经济情景的概率，并估计每种情景下一只股票获得正收益或负收益的概率。这些情景最适合用哪种方式表示?

A. 矩形树图。

B. 树状概率图。

C. 概率密度函数。

答案 / Answer: B

1. 先分经济情景，再分情景下的正负收益，形成条件分支，因此使用树状概率图。

Branch first by economic scenario and then by positive or negative return conditional on that scenario, so use a probability tree.

2. 矩形树图主要显示类别占比；概率密度函数描述连续随机变量，不适合本题的离散情景分支。

A treemap mainly shows categorical proportions, while a probability density function describes a continuous variable rather than these discrete scenario branches.

易错点 / Common trap

tree-map 和 tree diagram 不是同一种图。

A treemap and a probability-tree diagram are different visualizations.

Q53 有利情景下的条件期望 / Conditional expectation in the favorable scenario

巩固练习 / Reinforcement • 原题 PDF p.125 / 章节 p.11 • 答案 PDF p.53

53. A tree diagram contains the following information about the dividend per share payable by a company under two scenarios:

Scenario	Probability of Scenario	Dividend per Share	Probability of Dividend
Favorable	0.60	$2.00	0.80
		$1.50	0.20
Unfavorable	0.40	$0.75	0.30
		$0.50	0.70

The expected dividend per share under the favorable scenario is *closest* to:

A. $1.14.
B. $1.37.
C. $1.90.

中文题干与选项 / Chinese translation

树状图给出公司在两种情景下支付每股股利的信息，如下表。有利情景下的预期每股股利最接近多少?

情景	情景概率	每股股利	该情景下股利概率
有利	0.60	$2.00	0.80
有利	0.60	$1.50	0.20
不利	0.40	$0.75	0.30
不利	0.40	$0.50	0.70

A. $1.14。

B. $1.37。

C. $1.90。

答案 / Answer: C

1. 题目已给定有利情景，只用该情景内的条件概率：E(D|有利)=0.80*$2.00+0.20*$1.50=$1.90。

The favorable scenario is given, so use its conditional probabilities only: E(D|favorable)=0.80*$2.00+0.20*$1.50=$1.90.

2. $1.14=0.60*$1.90，多乘了有利情景概率；$1.37 是把有利与不利情景合并后的无条件期望。

$1.14=0.60*$1.90 incorrectly adds the favorable-scenario probability. $1.37 is the unconditional mean combining both scenarios.

3. 核对无条件值：0.60*1.90+0.40*(0.30*0.75+0.70*0.50)=1.37，但这不是本题所问。

For comparison, 0.60*1.90+0.40*(0.30*0.75+0.70*0.50)=1.37, but that is not the conditional expectation requested.

易错点 / Common trap

看到 under the favorable scenario，先把有利情景视为已发生。

When asked under the favorable scenario, treat that scenario as already given.

资料提示 / Source note

答案页文字说明中曾把第一个股利写成 $2.50，但原题及答案页公式均为 $2.00。本讲义按原题 $2.00 计算，并保留此勘误说明。

The answer prose once labels the first dividend $2.50, but the original question and answer formulas use $2.00. This guide uses the original $2.00 and records the typo.

4.2 期望、方差、标准差 / Expectation, variance and standard deviation

先学知识点 / Learn the concept first

期望值是概率加权平均。方差是相对于期望值的平方偏差的概率加权平均；标准差再开平方。

Expectation is a probability-weighted mean. Variance is the probability-weighted mean of squared deviations from expectation; standard deviation is its square root.

题目给出完整概率分布时，不是把几个可能结果当作样本再除以 n-1。先确认所有概率加总为 1。

When a full probability distribution is given, do not treat its possible outcomes as a sample and divide by n-1. First check that the probabilities sum to one.

E(X) = sum(p_i*x_i); Var(X) = sum[p_i*(x_i-E(X))^2]; SD(X) = sqrt(Var(X))

若 X 的单位是美元，方差单位为美元平方，标准差单位为美元。

If X is measured in dollars, variance is measured in dollars squared and standard deviation in dollars.

Q42 等概率情景的方差 / Variance of equally likely outcomes

巩固练习 / Reinforcement • 原题 PDF p.123 / 章节 p.9 • 答案 PDF p.42

42. An analyst assumes that a company's future EPS will be either $2.00, $2.20, or $2.40. If each scenario is equally *likely*, the variance [in $] of the company's future EPS is *closest* to:
    A. 0.03.
    B. 0.16.
    C. 0.20.

中文题干与选项 / Chinese translation

分析师假设公司未来 EPS 为 $2.00、$2.20 或 $2.40，且各情景等可能。未来 EPS 的方差［原文单位为 $］最接近多少?

A. 0.03。

B. 0.16。

C. 0.20。

答案 / Answer: A

1. E(EPS)=(2.00+2.20+2.40)/3=2.20。

E(EPS)=(2.00+2.20+2.40)/3=2.20.

2. Var(EPS)=[(2.00-2.20)^2+(2.20-2.20)^2+(2.40-2.20)^2]/3=0.026667，最接近 0.03。

Var(EPS)=[(2.00-2.20)^2+(2.20-2.20)^2+(2.40-2.20)^2]/3=0.026667, closest to 0.03.

3. 0.16 约等于标准差 sqrt(0.026667)，题目要求方差，所以不要开平方。

0.16 approximates the standard deviation sqrt(0.026667); the question asks for variance, so do not take the square root.

易错点 / Common trap

等可能的三种未来情景是概率分布，不是用 n-1 的三项样本。

Three equally likely future outcomes form a probability distribution, not a three-observation sample requiring n-1.

资料提示 / Source note

原题把方差单位写为 $；正确单位应为 $²。原答案页计算处使用 $²，数值答案 A 不变。

The stem labels variance in dollars; the correct unit is dollars squared. The answer's calculation uses squared-dollar units, and the numerical answer remains A.

Q52 非等概率情景的标准差 / Standard deviation with unequal probabilities

核心例题 / Core example • 原题 PDF p.125 / 章节 p.11 • 答案 PDF p.52

52. A discrete random variable X has the following probability distribution:

Probability	Outcome
0.20	35
0.30	50
0.50	80

The standard deviation of X is *closest* to:

A. 18.73.

B. 20.00.

C. 22.91.

中文题干与选项 / Chinese translation

离散随机变量 X 具有下表概率分布。X 的标准差最接近多少?

概率	结果
0.20	35
0.30	50
0.50	80

A. 18.73。

B. 20.00。

C. 22.91。

答案 / Answer: A

1. 先按概率求期望：E(X)=0.20*35+0.30*50+0.50*80=62。

First compute the probability-weighted mean: E(X)=0.20*35+0.30*50+0.50*80=62.

2. Var(X)=0.20*(35-62)^2+0.30*(50-62)^2+0.50*(80-62)^2=351。

Var(X)=0.20*(35-62)^2+0.30*(50-62)^2+0.50*(80-62)^2=351.

3. SD(X)=sqrt(351)=18.73499，最接近 18.73。

SD(X)=sqrt(351)=18.73499, closest to 18.73.

易错点 / Common trap

三个结果概率不同，不能先取 (35+50+80)/3，也不能忽略概率改算样本标准差。

The outcomes are not equally likely: do not use (35+50+80)/3 or replace probability weighting with sample standard deviation.

4.3 联合分布与协方差 / Joint distributions and covariance

先学知识点 / Learn the concept first

协方差衡量两变量相对各自均值的偏离是否同向。两者都高于均值或都低于均值时，偏差乘积为正。

Covariance measures whether deviations from each variable's own mean move together. Deviations have a positive product when both variables are above or both are below their means.

联合概率表中的每一个格子代表一对结果及其概率；概率为零的格子可以直接略去。

Each cell in a joint probability table represents a paired outcome and its probability. Cells with zero probability contribute nothing.

Cov(X,Y) = sum_i sum_j p_ij*(x_i-E(X))*(y_j-E(Y))

若用百分数数值代入，结果是“百分数平方”；若用小数收益率，必须一致地全部改成小数。

Using percentage-point inputs gives percent-squared units. If using decimal returns, convert every return consistently.

Q57 由联合概率计算协方差 / Covariance from joint probabilities

核心例题 / Core example • 原题 PDF p.126 / 章节 p.12 • 答案 PDF p.57

57. An analyst produces the following joint probability function for the returns on two companies, *X* and *Y:*

	Return of *Y*		
	Return of *Y*=15%	Return of *Y*=10%	Return of *Y*=5%
Return of *X*=20%	0.2	0	0
Return of *X*=15%	0	0.4	0
Return of *X*=10%	0	0	0.4

The expected returns of companies X and Y are 14% and 9%, respectively. The covariance of returns between X and Y (in percent squared) is *closest* to:

A. 0.

B. 5.

C. 14.

中文题干与选项 / Chinese translation

分析师给出 X 与 Y 两家公司收益率的联合概率如下表。X、Y 的预期收益分别为 14% 和 9%。两者收益率协方差（以百分数平方表示）最接近多少？

X 收益率 / Y 收益率	Y=15%	Y=10%	Y=5%
X=20%	0.2	0	0
X=15%	0	0.4	0
X=10%	0	0	0.4

A. 0。

B. 5。

C. 14。

答案 / Answer: C

1. 只取三个非零概率格，并对两变量分别减去自己的均值。

Use the three non-zero-probability cells and subtract each variable's own mean.

2. Cov=0.2*(20-14)*(15-9)+0.4*(15-14)*(10-9)+0.4*(10-14)*(5-9)=7.2+0.4+6.4=14。

Cov=0.2*(20-14)*(15-9)+0.4*(15-14)*(10-9)+0.4*(10-14)*(5-9)=7.2+0.4+6.4=14.

3. 使用百分数数值时结果是 14 个“百分数平方”；如果全部改用小数收益率，协方差为 0.0014。

Percentage-point inputs produce 14 in percent-squared units. Using decimal returns throughout instead gives covariance 0.0014.

易错点 / Common trap

把 X 收益减 Y 收益得到的是差额，不是协方差。

Subtracting Y's return from X's return gives a difference, not covariance.

单元 05 组合风险、安全第一与分布

Portfolio Risk, Safety First and Distributions

把概率统计连到投资问题：组合风险要计入相关性；安全第一关注跌破最低收益门槛；价格与连续复利收益使用不同分布。

Connect statistics to investment decisions: portfolio risk incorporates correlation, safety first focuses on falling below a minimum return, and prices and continuously compounded returns use different distributions.

本单元 5 题 / 5 questions in this unit

单元速记 / Unit essentials

零相关下组合标准差仍大于零；将组合方差与标准差分清。

Zero correlation does not make portfolio standard deviation zero; distinguish variance from standard deviation.

安全第一：减最低目标，再除标准差；不要只选收益最高或波动最低。

Safety first: subtract the minimum target, then divide by standard deviation; highest return or lowest volatility alone is insufficient.

连续复利收益正态，对应价格对数正态。

Normal log returns imply lognormal prices.

5.1 两资产组合的标准差 / Standard deviation of a two-asset portfolio

先学知识点 / Learn the concept first

组合收益是权重加权平均，但组合标准差通常不能直接对单个资产标准差取加权平均。先求方差，再开平方。

Portfolio return is a weighted average, but portfolio standard deviation generally is not a weighted average of asset standard deviations. Compute variance first, then take the square root.

当相关系数为零，协方差项为零，单项风险仍然存在。分散化减少风险不等于消除所有风险。

When correlation is zero, the covariance term disappears, but individual variance terms remain. Diversification reduces risk without necessarily eliminating it.

sigma_p^2 = w1^2*sigma1^2 + w2^2*sigma2^2 + 2*w1*w2*sigma1*sigma2*rho12

权重必须平方；最后别忘记对组合方差开平方。

Square the weights and take the square root of portfolio variance at the end.

Q62 零相关两资产组合风险 / Risk of two uncorrelated assets

核心例题 / Core example • 原题 PDF p.127 / 章节 p.13 • 答案 PDF p.62

62. An equally weighted portfolio consists of two securities, each with a standard deviation of 3%. If the two securities' returns are uncorrelated, the portfolio's standard deviation is *closest* to:
    A. 0.0%.
    B. 2.1%.
    C. 3.0%.

中文题干与选项 / Chinese translation

一个等权组合包含两只证券，每只证券的标准差均为 3%。若两只证券的收益率不相关，组合标准差最接近多少？

A. 0.0%。

B. 2.1%。

C. 3.0%。

答案 / Answer: B

1. 等权表示每只权重 0.5，零相关使协方差项等于零。

Equal weighting gives each asset weight 0.5, and zero correlation removes the covariance term.

2. 以百分数数值计算：sigma_p^2=0.5^2*3^2+0.5^2*3^2=4.5；sigma_p=sqrt(4.5)=2.1213%，最接近 2.1%。

Using percentage-point inputs: sigma_p^2=0.5^2*3^2+0.5^2*3^2=4.5; sigma_p=sqrt(4.5)=2.1213%, closest to 2.1%.

易错点 / Common trap

零相关仍有单项风险；3% 是直接平均标准差得到的错误结果。

Zero correlation leaves individual variance risk; 3% incorrectly averages the standard deviations.

5.2 Roy 安全第一准则 / Roy's safety-first criterion

先学知识点 / Learn the concept first

短缺风险是组合收益低于最低可接受收益 R_L 的概率。正态收益假设下，最大化安全第一比率等价于最小化这一概率。

Shortfall risk is the probability that portfolio return falls below a minimum acceptable return R_L. With normally distributed returns, maximizing the safety-first ratio minimizes this probability.

先把需要提取的金额除以本金，得到收益门槛；再把每个组合高于门槛的预期收益除以其标准差。

First divide the required withdrawal by initial capital to obtain the return threshold. Then divide each portfolio's expected return above that threshold by its standard deviation.

该准则关注下行事件，但公式的分母仍是完整标准差；它不使用半标准差。常见题目假设收益率正态，并非价格正态。

The criterion focuses on a downside event, but its denominator is full standard deviation, not semideviation. These questions assume normal returns, not normally distributed prices.

SFR = (E(R_p)-R_L)/sigma_p; P(R_p<R_L) = Phi(-SFR)

比率越大，在正态收益假设下跌破门槛的概率越低。Phi 为标准正态累积分布函数。

A larger ratio means a lower shortfall probability under normal returns. Phi is the standard normal cumulative distribution function.

Q32 安全第一准则的风险目标 / Risk objective of the safety-first criterion

巩固练习 / Reinforcement • 原题 PDF p.120 / 章节 p.6 • 答案 PDF p.32

32. Roy's safety-first criterion:
    A. evaluates only downside risk.
    B. uses semideviation as a risk measure.
    C. assumes asset prices are normally distributed.

中文题干与选项 / Chinese translation

Roy 安全第一准则：

A. 只评价下行风险。

B. 使用半标准差作为风险度量。

C. 假设资产价格服从正态分布。

答案 / Answer: A

1. 它关注组合收益低于最低可接受门槛的短缺风险，因此目标是下行事件，A 正确。

It focuses on shortfall risk, the chance that portfolio return falls below a minimum acceptable threshold, so its objective concerns a downside event.

2. SFR=(预期收益-门槛)/标准差，分母不是半标准差；本题使用收益率正态假设，而不是价格正态。

SFR=(expected return-threshold)/standard deviation, not semideviation. The relevant normality assumption concerns returns, not asset prices.

易错点 / Common trap

“关注下行风险”不等于“公式分母必须用下行偏差”。

Focusing on downside risk does not mean the denominator must be downside deviation.

Q19 安全第一比率与缺失表格 / Safety-first ratios with an incomplete source table

巩固练习 / Reinforcement • 原题 PDF p.118 / 章节 p.4 • 答案 PDF p.19

19. A portfolio manager will invest $100,000 and is presented with the following information about three portfolios with normally distributed returns:

Expected Annual	Return Standard	Deviation of Returns
Portfolio	1	23%
Portfolio	2	12%
Portfolio	3	15%

If the manager wants to withdraw $5,000 in one year without invading initial capital, the safety-first optimal portfolio is:

A. Portfolio 1.
B. Portfolio 2.
C. Portfolio 3.

中文题干与选项 / Chinese translation

组合经理将投资 $100,000，并取得三个收益率正态分布的组合资料（原题表格错排，部分信息缺失，见表）。若希望一年后提取 $5,000 且不侵蚀本金，安全第一最优组合是哪一个?

原表第 1 列	原表第 2 列	原表第 3 列
组合	1	23%
组合	2	12%
组合	3	15%
标准差对应数据	待补充	Missing information

A. 组合 1。

B. 组合 2。

C. 组合 3。

答案 C（依答案页补充数据）/ C (using data from the solution page)

1. 仅凭原题错排的表格，缺少可用标准差，无法独立计算。答案文件第 19 页明确列出预期收益 23%、12%、15%，以及标准差 15%、6%、8%；以下仅演示该答案文件的数据。

The malformed question table lacks usable standard deviations, so it is insufficient by itself. Answer page 19 explicitly supplies expected returns 23%, 12%, 15% and standard deviations 15%, 6%, 8%. The following demonstration uses those answer-source figures.

2. 最低收益门槛 R_L=$5,000/$100,000=5%。

The minimum acceptable return is R_L=$5,000/$100,000=5%.

3. 按答案文件数据：SFR1=(23%-5%)/15%=1.20；SFR2=(12%-5%)/6%=1.1667；SFR3=(15%-5%)/8%=1.25。最高为组合 3，故原答案 C。

Using the answer-source figures: SFR1=(23%-5%)/15%=1.20; SFR2=(12%-5%)/6%=1.1667; SFR3=(15%-5%)/8%=1.25. Portfolio 3 is highest, matching supplied answer C.

易错点 / Common trap

不应只选预期收益最高的组合 1，或标准差最低的组合 2；必须比较减去门槛后的比率。

Do not choose Portfolio 1 merely for its highest expected return or Portfolio 2 for lowest volatility; compare returns above the threshold per unit of risk.

资料提示 / Source note

原题表格的标题和列错排；标准差为“待补充 / Missing information”。本讲义未把答案中出现的数据冒充为完整原题，答案 C 以答案文件所列数据为条件。原答案也把题目美元符号写成欧元，比例计算不受影响。

The original table has malformed headers and columns; standard deviations are Missing information. This guide does not present answer-source figures as if they appeared in the original question. C is conditional on those figures. The answer also switches the dollar symbol to euros, which does not affect the ratio.

5.3 连续复利收益与对数正态价格 / Log returns and lognormal prices

先学知识点 / Learn the concept first

连续复利收益 r=ln(P_T/P_0)。若 r 正态，P_T=P_0*exp(r) 就是对数正态。固定初始价格并取指数后，价格为正。

Continuously compounded return is r=ln(P_T/P_0). If r is normal, P_T=P_0*exp(r) is lognormal. Exponentiation with a fixed positive initial price gives a positive price.

对数正态分布不对称、右偏，下界为零，没有有限上界。它的均值不同于对应正态变量的均值。

A lognormal distribution is asymmetric and right-skewed, bounded below by zero and without a finite upper bound. Its mean differs from the mean of its underlying normal variable.

> r = ln(P_T/P_0); P_T = P_0*exp(r)
>
> 先辨认题目讨论的是收益率，还是价格。
>
> First identify whether the question concerns returns or prices.

Q72 正态对数收益与对数正态价格 / Normal log returns and lognormal prices

巩固练习 / Reinforcement • 原题 PDF p.129 / 章节 p.15 • 答案 PDF p.72

72. If a stock's continuously compounded return is normally distributed, the future stock price is *most likely*.
    A. normally distributed.
    B. uniformly distributed.
    C. lognormally distributed.

中文题干与选项 / Chinese translation

如果一只股票的连续复利收益服从正态分布，其未来股价最可能服从：

A. 正态分布。

B. 均匀分布。

C. 对数正态分布。

答案 / Answer: C

1. 连续复利收益 r=ln(P_T/P_0)，所以 P_T=P_0*exp(r)。

Continuously compounded return is r=ln(P_T/P_0), so P_T=P_0*exp(r).

2. 正态随机变量取指数后服从对数正态分布；固定的正初始价格乘在前面不改变这一分布类型。

Exponentiating a normal random variable produces a lognormal variable; multiplying by a fixed positive initial price preserves that distribution family.

易错点 / Common trap

收益率与价格不是同一个随机变量，不能直接套相同分布。

Returns and prices are different random variables and need not share the same distribution.

Q75 对数正态分布的形状 / Shape of a lognormal distribution

巩固练习 / Reinforcement • 原题 PDF p.130 / 章节 p.16 • 答案 PDF p.75

75. The lognormal distribution:
    A. is unbounded.
    B. is asymmetrical.
    C. has the same mean as that of its associated normal distribution.

中文题干与选项 / Chinese translation

对数正态分布：

A. 没有边界。

B. 是不对称的。

C. 其均值与对应正态分布的均值相同。

答案 / Answer: B

1. 对数正态分布是右偏的不对称分布，因此选 B。

A lognormal distribution is asymmetric and right-skewed, so B is correct.

2. 它有零下界而没有有限上界，所以 A 所说完全无界不准确；对数变换前后的均值也不会一般相等。

It is bounded below by zero but has no finite upper bound, so A is not accurate as a claim of no bounds. Means before and after the logarithmic transformation are not generally equal.

易错点 / Common trap

lognormal 表示“取对数后正态”，不是“自身对称正态”。

Lognormal means normal after taking logs, not symmetric and normal in its original scale.

单元 06 抽样与估计

Sampling and Estimation

先分清总体、样本和统计量，再判断抽样方式、标准误与置信区间。核心问题是：用一份样本推断总体时，估计有多不确定?

Distinguish the population, sample, and statistic before choosing a sampling method, standard error, or confidence interval. The central question is how uncertain a population estimate is when it comes from one sample.

本单元 10 题 / 10 questions in this unit

单元速记 / Unit essentials

置信区间用标准误，不用实际抽样误差。

Confidence intervals use standard error, not realized sampling error.

分层：每层都抽；整群：随机选部分群；Bootstrap：原样本内有放回重抽。

Stratified: sample every stratum. Cluster: select some clusters. Bootstrap: resample with replacement from the original sample.

中心极限定理说的是样本均值的分布，不是把原始总体变成正态。

The central limit theorem concerns the distribution of the sample mean; it does not make the original population normal.

6.1 抽样方式：每层都抽，还是只抽部分群？ / Sampling designs: sample every stratum or only selected clusters?

先学知识点 / Learn the concept first

概率抽样让每个总体成员具有已知且非零的入样概率；在其他条件相同的情况下，通常比非概率抽样更有代表性，但不保证某一个样本必然代表总体。

Probability sampling gives each population member a known, nonzero inclusion probability. Other things equal, it usually improves representativeness relative to non-probability sampling, but it does not guarantee that every realized sample is representative.

分层随机抽样：先按行业等特征分层，再从每一层随机抽取样本；按总体占比分配样本可以保留总体结构。

Stratified random sampling divides the population by characteristics such as sector and randomly samples within every stratum. Proportional allocation preserves the population structure.

整群抽样：把总体分成群，随机选部分群，再调查所选群的全部成员或继续抽样。市场调查常因地域和成本采用此方法。

Cluster sampling divides the population into clusters, selects some clusters at random, and then surveys all or a sample of members within the selected clusters. Geographic and cost considerations often make this useful in market surveys.

按公司名称首字母选择样本并没有随机选择机制，属于非概率抽样。系统抽样则通常从随机起点开始，每隔固定数量选择一个成员。

Selecting firms by the first letter of their names has no random selection mechanism and is non-probability sampling. Systematic sampling typically starts at a random point and selects every kth member.

Q33 概率抽样与代表性 / Probability sampling and representativeness

巩固练习 / Reinforcement • 原题 PDF p.121 / 章节 p.7 • 答案 PDF p.33

33. All else being equal, when compared to non-probability sampling, probability sampling *most likely* yields:
    A. a less representative sample.
    B. an equally representative sample.
    C. a more representative sample.

中文题干与选项 / Chinese translation

在其他条件相同的情况下，与非概率抽样相比，概率抽样最可能得到：

A. 代表性更差的样本。

B. 代表性相同的样本。

C. 代表性更好的样本。

答案 / Answer: C

1. 概率抽样通过随机机制和已知入样概率减少主观选择偏差。

Probability sampling uses a random mechanism and known inclusion probabilities to reduce subjective selection bias.

2. 因此在题设“其他条件相同”下，样本通常更有代表性；这是一种通常的比较结果，不是对每份样本的保证。

Therefore, under the question’s all-else-equal condition, it usually produces a more representative sample. This is a general comparison, not a guarantee for every sample.

易错点 / Common trap

“更有代表性”不等于没有抽样误差。

More representative does not mean free from sampling error.

资料提示 / Source note

原解析把概率抽样概括成每个成员入选概率相同，表述过窄。更一般的要求是入选概率已知且非零，并不一定相等；C答案不变。

The supplied explanation describes all probability sampling as giving members equal selection probabilities, which is too narrow. The general requirement is known, nonzero inclusion probabilities, which need not be equal. Answer C is unchanged.

Q92 分层随机抽样 / Stratified random sampling

巩固练习 / Reinforcement • 原题 PDF p.133 / 章节 p.19 • 答案 PDF p.92

92. Grouping all publicly traded US firms by sector and then randomly selecting subsamples of firms from each sector according to the sector's proportion in the total population is an example of:
    A. cluster sampling.
    B. simple random sampling.
    C. stratified random sampling.

中文题干与选项 / Chinese translation

把所有美国上市公司按行业分组，然后按照各行业占总体的比例，从每个行业随机抽取公司子样本。这是：

A. 整群抽样。

B. 简单随机抽样。

C. 分层随机抽样。

答案 / Answer: C

1. “按行业分组”是分层；“每个行业都随机抽取”表明不是只选部分群。

Grouping by sector creates strata; randomly sampling from every sector distinguishes this from selecting only some clusters.

2. 样本按行业在总体中的占比配置，属于按比例分配的分层随机抽样。

Allocating the subsamples according to sector population shares makes this proportionate stratified random sampling.

易错点 / Common trap

看到“分组”先不要选整群，还要看是每组抽人还是抽部分整组。

Do not choose cluster sampling merely because groups are mentioned; check whether every group is sampled or only some groups are selected.

Q87 整群抽样的应用 / Applications of cluster sampling

巩固练习 / Reinforcement • 原题 PDF p.132 / 章节 p.18 • 答案 PDF p.87

87. In which of the following cases is cluster sampling *most likely* used? When:
    A. conducting a market survey.
    B. auditing financial statements.
    C. creating a bond portfolio to mirror the performance of a specified index.

中文题干与选项 / Chinese translation

以下哪一种情形最可能采用整群抽样?

A. 开展市场调查。

B. 审计财务报表。

C. 建立一个复制指定指数表现的债券组合。

答案 / Answer: A

1. 市场调查常面对分散在不同地点的大量对象，先选部分地区或群组可以减少调查成本。

Market surveys often cover many people spread across locations, so selecting some geographic clusters can reduce cost.

2. 复制债券指数更典型的是分层抽样，在期限、信用等层中选债以维持指数特征。题目比较“最可能”，并非声称其他场景绝不能用整群抽样。

Replicating a bond index more typically uses stratified sampling across maturity, credit, or similar categories to preserve index characteristics. The question asks most likely, not whether cluster sampling is impossible elsewhere.

易错点 / Common trap

分层希望保留每层比例；整群更强调抽取部分群的调查便利和成本。

Stratification preserves the representation of every stratum; clustering often emphasizes the practicality and cost of surveying selected groups.

Q73 非概率抽样 / Non-probability sampling

巩固练习 / Reinforcement • 原题 PDF p.129 / 章节 p.15 • 答案 PDF p.73

73. An analyst considers the population of all existing stocks and selects those where the company name starts with the letter P. This sampling procedure is *most likely* an example of.
    A. systematic sampling.
    B. non-probability sampling.
    C. two-stage cluster sampling.

中文题干与选项 / Chinese translation

分析师把所有现存股票作为总体，并选择公司名称以字母P开头的股票。该抽样过程最可能属于：

A. 系统抽样。

B. 非概率抽样。

C. 两阶段整群抽样。

答案 / Answer: B

1. 按名称首字母P作确定性筛选，没有给每家公司一个随机入选机会。

A deterministic filter for names starting with P gives firms no random selection opportunity.

2. 系统抽样应按固定间隔抽取；两阶段整群抽样则先随机选群、再在群内抽样。题干都没有这些步骤。

Systematic sampling selects at a fixed interval, while two-stage cluster sampling randomly selects clusters and then samples within them. Neither process appears in the question.

易错点 / Common trap

有规则的选取，不一定是统计学上的systematic sampling。

A rule-based selection is not automatically systematic sampling in the statistical sense.

6.2 抽样误差、标准误与中心极限定理 / Sampling error, standard error, and the central limit theorem

先学知识点 / Learn the concept first

抽样误差是某次样本统计量与它要估计的总体参数之间的差；标准误是该统计量在重复抽样中的标准差。两者不能互换。

Sampling error is the difference between a realized sample statistic and the population parameter it estimates. Standard error is the standard deviation of that statistic across repeated samples. They are not interchangeable.

在通常的独立同分布、有限方差条件下，样本足够大时，样本均值的抽样分布近似正态；总体本身不需要正态。

Under the usual independent and identically distributed sampling conditions with finite variance, the sampling distribution of the sample mean becomes approximately normal for sufficiently large samples. The population itself need not be normal.

一致性表示样本量增大时估计量趋近真实参数。样本均值的一致性通常由大数定律说明；它与本题使用的中心极限定理条件相容。

Consistency means that an estimator converges to the true parameter as sample size grows. The law of large numbers most directly establishes consistency of the sample mean; this is compatible with the conditions used for the central limit theorem here.

Sampling error = x_bar - mu

样本均值减总体均值；总体均值未知时，实际误差通常不可直接观测。

The sample mean minus the population mean; the realized error is usually not directly observable when the population mean is unknown.

SE(x_bar) = sigma / sqrt(n); estimated SE = s / sqrt(n)

总体标准差已知用 sigma，未知时通常以样本标准差 s 估计。

Use sigma when the population standard deviation is known; otherwise it is commonly estimated by the sample standard deviation s.

Q40 抽样误差定义 / Definition of sampling error

巩固练习 / Reinforcement • 原题 PDF p.122 / 章节 p.8 • 答案 PDF p.40

40. Sampling error is the difference between the observed value of a:
    A. random variable and the respective statistic.
    B. random variable and its hypothesized value.
    C. statistic and the quantity it is intended to estimate.

中文题干与选项 / Chinese translation

抽样误差是以下哪两者的差?

A. 某随机变量的观测值与相应统计量。

B. 某随机变量的观测值与其假设值。

C. 某统计量的观测值与它打算估计的数值。

答案 / Answer: C

1. 统计量来自样本，参数描述总体；抽样误差比较的是这两者。

A statistic comes from a sample and a parameter describes the population. Sampling error compares the two.

2. 例如样本均值的抽样误差写成x_bar - mu，方向为估计值减真实参数。

For the sample mean, sampling error is x_bar - mu: the estimate minus the true parameter.

易错点 / Common trap

抽样误差不是样本内部各个观测值围绕样本均值的偏差。

Sampling error is not each individual observation’ s deviation from the sample mean.

Q41 中心极限定理与一致性 / The central limit theorem and consistency

巩固练习 / Reinforcement • 原题 PDF p.122 / 章节 p.8 • 答案 PDF p.41

41. The central limit theorem:
    A. requires that the population be approximately normally distributed.
    B. implies that the sample mean is a consistent estimator of the population mean.
    C. states that the product of independent random variables is normally distributed.

中文题干与选项 / Chinese translation

关于中心极限定理，以下哪项表述正确?

A. 它要求总体近似服从正态分布。

B. 它意味着样本均值是总体均值的一致估计量。

C. 它指出独立随机变量的乘积服从正态分布。

答案 / Answer: B

1. 原答案选B。在通常的独立同分布、有限方差条件下，样本均值的方差是sigma^2/n，样本量增加时估计更集中于总体均值。

The supplied answer is B. Under the usual IID finite-variance conditions, the sample mean has variance sigma^2/n, so the estimate concentrates around the population mean as the sample grows.

2. A错误：总体不必正态。C错误：中心极限定理讨论标准化的和或均值的极限分布，不是一般独立变量乘积。

A is wrong because the population need not be normal. C is wrong because the theorem concerns the limiting distribution of standardized sums or means, not a general product of independent variables.

易错点 / Common trap

不要把“样本均值近似正态”误记为“总体近似正态”。

Do not confuse approximate normality of the sample mean with normality of the population.

资料提示 / Source note

严格概念上，一致性通常直接由大数定律说明。这里保留原题B，同时补足其适用的常规条件。

Strictly, consistency is most directly established by the law of large numbers. B is retained as the supplied answer with its usual conditions made explicit.

Q88 非正态总体的样本均值 / Sample means from a non-normal population

巩固练习 / Reinforcement • 原题 PDF p.132 / 章节 p.18 • 答案 PDF p.88

88. Samples are drawn from a population that follows a binomial distribution with a probability of success on a trial of 0.3. According to the central limit theorem, as the sample size increases, the distribution of the sample mean approaches a
A. negatively skewed distribution.
B. Symmetric distribution.
C. positively skewed distribution.

中文题干与选项 / Chinese translation

从一个每次试验成功概率为0.3的二项分布总体中抽取样本。依据中心极限定理，随着样本量增加，样本均值的分布趋近于：

A. 负偏分布。

B. 对称分布。

C. 正偏分布。

答案 / Answer: B

1. 二项分布的具体偏度不决定大样本均值分布的最终形状；在中心极限定理条件下，均值的抽样分布近似正态。

The original binomial distribution’ s skew does not determine the eventual shape of the large-sample mean distribution. Under the CLT conditions, the sampling distribution of the mean is approximately normal.

2. 正态分布是对称的，所以B正确。这里是增加样本中的观测数，而不是把总体每次试验成功概率改成0.5。

The normal distribution is symmetric, so B is correct. The sample size increases; the population success probability is not changed to 0.5.

易错点 / Common trap

区分原始二项分布与样本均值的抽样分布。

Distinguish the original binomial distribution from the sampling distribution of its sample mean.

6.3 置信区间：点估计加减误差界限 / Confidence intervals: estimate plus or minus a margin of error

先学知识点 / Learn the concept first

置信区间由点估计、可靠性系数（例如 t 或 z 临界值）和标准误构成。不要把标准误误读为某次实际抽样误差。

A confidence interval combines a point estimate, a reliability factor such as a t or z critical value, and a standard error. Do not substitute the realized sampling error for the standard error.

95%置信水平描述的是重复抽样构造区间的方法：长期看约95%的这类区间覆盖固定的总体参数。

A 95% confidence level describes the interval-construction procedure: over repeated samples, about 95% of such intervals would cover the fixed population parameter.

> Confidence interval = point estimate +/- critical value x standard error
>
> 误差界限为临界值乘标准误。
>
> The margin of error is the critical value multiplied by the standard error.

Q01 置信区间的构成 / Components of a confidence interval

核心例题 / Core example • 原题 PDF p.115 / 章节 p.1 • 答案 PDF p.1

1. Which of the following factors is not used in the calculation of a confidence interval?
   A. Point estimate.
   B. Sampling error.
   C. Reliability factor.

中文题干与选项 / Chinese translation

以下哪一项不用于计算置信区间?

A. 点估计。

B. 抽样误差。

C. 可靠性系数。

答案 / Answer: B

1. 公式：置信区间 = 点估计 ± 可靠性系数 × 标准误。

Formula: confidence interval = point estimate ± reliability factor × standard error.

2. A是区间中心，C决定区间宽度，因此都要用。B指统计量与真实参数之间某次实际的差，不是公式中的标准误。

A provides the center and C helps determine the width, so both are required. B is a realized difference between a statistic and the true parameter, not the standard error in the formula.

易错点 / Common trap

看到sampling error不要自动当作standard error。

Do not read sampling error as standard error.

6.4 Bootstrap：从原样本反复有放回重抽样 / Bootstrap: repeatedly resample with replacement from the original sample

先学知识点 / Learn the concept first

Bootstrap把现有样本当作总体的经验近似，反复有放回地抽取通常与原样本同样大小的重样本，并计算每个重样本的目标统计量。

The bootstrap treats the observed sample as an empirical approximation to the population, repeatedly draws samples with replacement, usually of the original sample size, and computes the statistic of interest for each resample.

估计样本均值的标准误时，先得到每个重样本的均值，再计算这些均值之间的标准差。

To estimate the standard error of a sample mean, calculate the mean of every resample and then the standard deviation of those resample means.

```
SE_boot = sqrt[ sum((mean_b - mean_of_bootstrap_means)^2) / (B - 1) ]
```

B是重抽样次数；这里计算的是重样本均值之间的离散程度。

B is the number of resamples; the calculation measures dispersion across the resample means.

Q24 Bootstrap识别 / Identifying bootstrap sampling

巩固练习 / Reinforcement · 原题 PDF p.119 / 章节 p.5 · 答案 PDF p.24

24. An analyst draws samples from an original sample to estimate the standard error of a population mean. Which of the following best describes this sampling procedure?
    A. Bootstrap method
    B. Cluster sampling method
    C. Convenience sampling method

中文题干与选项 / Chinese translation

分析师从一份原始样本中再次抽取样本，用来估计总体均值估计量的标准误。以下哪一项最能描述这一抽样过程？

A. Bootstrap自助法。

B. 整群抽样法。

C. 便利抽样法。

答案 / Answer: A

1. 题干的关键信息是“从原始样本中再抽样”；这对应Bootstrap。

The key phrase is resampling from an original sample, which identifies the bootstrap.

2. 完整过程是有放回地反复重抽样，计算每份重样本的均值，再看这些均值的离散程度。整群抽样是从总体的群组中选群。

The full procedure repeatedly resamples with replacement, computes each resample mean, and measures the dispersion of those means. Cluster sampling instead selects groups from the population.

易错点 / Common trap

从原样本再抽样与从总体重新抽样不是同一件事。

Resampling from the original sample differs from drawing a new sample directly from the population.

Q51 Bootstrap标准误 / Bootstrap standard error

核心例题 / Core example · 原题 PDF p.124 / 章节 p.10 · 答案 PDF p.51

51. Which of the following is required to compute the standard error of a sample mean using the bootstrap resampling method?
    A. The mean of each resample
    B. The mean of the original sample
    C. The standard deviation of the original sample

中文题干与选项 / Chinese translation

用Bootstrap重抽样法计算样本均值的标准误时，需要以下哪一项？

A. 每个重样本的均值。

B. 原始样本的均值。

C. 原始样本的标准差。

答案 / Answer: A

1. 每一次重抽样都计算一个均值，形成一组Bootstrap均值。

Calculate one mean per resample to form a collection of bootstrap means.

2. 这些均值的样本标准差就是Bootstrap标准误的估计。仅有原样本均值或标准差，并不足以执行这项重抽样计算。

The sample standard deviation of those means estimates the bootstrap standard error. The original sample mean or standard deviation alone is not sufficient for this resampling calculation.

易错点 / Common trap

要计算“均值之间的标准差”，不是把每份重样本的标准差简单平均。

Compute the standard deviation across means, not the average of the resamples’ standard deviations.

单元 07 假设检验

Hypothesis Testing

做题先判断检验对象和样本关系，再选统计量与自由度。最容易混淆的是配对样本、独立样本、相关系数检验，以及第一类和第二类错误。

First identify the parameter being tested and how the samples are related; then choose the statistic and degrees of freedom. The main traps are paired versus independent samples, correlation tests, and Type I versus Type II errors.

本单元 12 题 / 12 questions in this unit

单元速记 / Unit essentials

配对均值差：n - 1；相关系数及简单回归残差：n - 2；独立等方差双样本均值差：n1 + n2 - 2。

Paired mean difference: n - 1. Correlation and simple-regression residuals: n - 2. Independent equal-variance two-sample mean difference: n1 + n2 - 2.

第一类：错拒真；第二类：未拒假；功效：正确拒绝假。

Type I: reject a true null. Type II: fail to reject a false null. Power: correctly reject a false null.

题目问least likely时先圈出否定方向，再在适用条件之间比较。

When the question asks least likely, mark the negative direction before comparing the conditions.

真实状态 / Truth	拒绝 H0 / Reject H0	不拒绝 H0 / Fail to reject H0
H0 为真 / H0 true	I 类错误 / Type I error 概率 / Probability: α	正确决定 / Correct decision
H0 为假 / H0 false	正确决定 / Correct decision 检验功效 / Power: 1-β	II 类错误 / Type II error 概率 / Probability: β

7.1 原假设、备择假设、错误与检验功效 / Null and alternative hypotheses, errors, and power

先学知识点 / Learn the concept first

备择假设表达希望得到证据支持的条件，例如总体均值大于零。原假设包括等号；右尾检验可写H0: mu <= 0，Ha: mu > 0，并在边界mu = 0计算统计量。

The alternative expresses the condition for which evidence is sought, such as a population mean greater than zero. The null includes equality: for a right-tailed test, H0: mu <= 0 and Ha: mu > 0, with the statistic evaluated at the boundary mu = 0.

第一类错误：拒绝真实的H0，概率为alpha。第二类错误：没有拒绝错误的H0，概率为beta。检验功效：正确拒绝错误的H0，等于1 - beta。

A Type I error rejects a true H0 and has probability alpha. A Type II error fails to reject a false H0 and has probability beta. Power is the probability of correctly rejecting a false H0, equal to 1 - beta.

p值是在H0成立时，得到至少同样极端结果的概率；p <= alpha时拒绝H0。它不是H0为真的概率。未拒绝H0不等于证明H0为真。

A p-value is the probability, assuming H0 is true, of a result at least as extreme as the observed one. Reject H0 when p <= alpha. It is not the probability that H0 is true. Failing to reject H0 does not prove it true.

Power = 1 - beta

beta是第二类错误的概率，不是第一类错误概率alpha。

Beta is the probability of a Type II error, not the Type I error probability alpha.

Q15 单尾备择假设 / One-sided alternative hypothesis

巩固练习 / Reinforcement · 原题 PDF p.117 / 章节 p.3 · 答案 PDF p.15

15. To test whether a population's mean, μ, is greater than zero, the alternative hypothesis should be formulated as:
    A. $\mu \leq 0$.
    B. $\mu \geq 0$.
    C. $\mu > 0$.

中文题干与选项 / Chinese translation

要检验总体均值mu是否大于零，备择假设应如何表述?

A. mu <= 0。

B. mu >= 0。

C. mu > 0。

答案 / Answer: C

1. 希望得到证据支持的是“均值大于零”，所以Ha: mu > 0。

The condition for which evidence is sought is a mean greater than zero, so Ha: mu > 0.

2. 对应H0: mu <= 0；等号放在原假设中，计算检验统计量时使用边界mu = 0。

The corresponding null is H0: mu <= 0. Equality belongs to the null, with the statistic evaluated at the boundary mu = 0.

易错点 / Common trap

greater than是严格大于，不是大于或等于。

Greater than is strict, not greater than or equal to.

Q36 检验功效 / Power of a test

巩固练习 / Reinforcement · 原题 PDF p.121 / 章节 p.7 · 答案 PDF p.36

36. The probability of correctly rejecting a null hypothesis is best defined as the:
    A. *p*-value.
    B. power of the test.
    C. level of significance.

中文题干与选项 / Chinese translation

正确拒绝原假设的概率，最恰当的定义是:

A. p值。

B. 检验功效。

C. 显著性水平。

答案 / Answer: B

1. 只有H0为假时，拒绝H0才是正确决定；这一概率叫检验功效。

Rejecting H0 is correct when H0 is false; this probability is the power of the test.

2. 功效 = 1 - beta。显著性水平alpha是拒绝真实H0的第一类错误概率，与功效不同。

Power = 1 - beta. The significance level alpha is the Type I error probability of rejecting a true H0, which is different from power.

易错点 / Common trap

correctly rejecting意味着拒绝的是假的H0。

Correctly rejecting means rejecting a false H0.

Q64 功效与第二类错误 / Power and Type II error

巩固练习 / Reinforcement • 原题 PDF p.128 / 章节 p.14 • 答案 PDF p.64

64. The probability of correctly rejecting a false null hypothesis is best described as one minus the:
    A. test statistic's p-value.
    B. probability of a Type I error.
    C. probability of a Type II error.

中文题干与选项 / Chinese translation

正确拒绝错误原假设的概率，最恰当地描述为1减去以下哪一项?

A. 检验统计量的p值。

B. 第一类错误的概率。

C. 第二类错误的概率。

答案 / Answer: C

1. 在H0实际上错误时，有两种结果：正确拒绝，或未能拒绝。后者是第二类错误beta。

When H0 is actually false, either it is correctly rejected or the test fails to reject it. The latter is a Type II error with probability beta.

2. 这两种结果互补，所以正确拒绝的概率为1 - beta。

The two outcomes are complementary, so the probability of correct rejection is 1 - beta.

易错点 / Common trap

1 - alpha不是检验功效；它对应H0为真时不犯第一类错误的概率。

1 - alpha is not power; it is the probability of avoiding a Type I error when H0 is true.

资料提示 / Source note

原解析把p值说成与第一类错误概率同义，这不准确。alpha是预先设定的错误率阈值，p值根据观测数据在H0下计算；两者并非同一数值。C答案正确。

The supplied explanation equates the p-value with the Type I error probability, which is inaccurate. Alpha is a prespecified error-rate threshold; the p-value is calculated from the observed data under H0. They are not the same quantity. Answer C is correct.

Q93 第二类错误 / Type II error

核心例题 / Core example • 原题 PDF p.133 / 章节 p.19 • 答案 PDF p.93

93. In hypothesis testing, which of the following best describes a Type II error?
    A. Rejecting a true null hypothesis.
    B. Rejecting a false null hypothesis.
    C. Failure to reject a false null hypothesis.

中文题干与选项 / Chinese translation

在假设检验中，以下哪一项最恰当地描述第二类错误?

A. 拒绝真实的原假设。

B. 拒绝错误的原假设。

C. 未能拒绝错误的原假设。

答案 / Answer: C

1. 第二类错误发生在H0为假，但检验没有拒绝H0。

A Type II error occurs when H0 is false but the test fails to reject it.

2. A是第一类错误，B是正确决定。C才是漏掉应被拒绝的错误H0。

A is a Type I error and B is a correct decision. C is the failure to reject a false H0 that should be rejected.

易错点 / Common trap

“未拒绝”不等于“接受为真”；这个措辞有助于避免把错误类型记反。

Failing to reject is not the same as accepting as true; this wording helps avoid reversing the error types.

7.2 均值检验：配对与独立样本要分开 / Mean tests: distinguish paired and independent samples

先学知识点 / Learn the concept first

同一月份两个指数的收益是一对观测。配对t检验先计算每对差值，再对这些差值做单样本均值检验，所以n对数据的自由度是n - 1。

Returns on two indexes in the same month form a pair. A paired t-test first calculates each pairwise difference, then tests the mean of those differences. With n pairs, the degrees of freedom are n - 1.

两个独立正态总体、方差未知但相等时，使用合并方差t检验，自由度为n1 + n2 - 2。改变假设均值差只改变统计量的分子，不改变样本和合并方差。

For independent normal populations with unknown but equal variances, use a pooled-variance t-test with n1 + n2 - 2 degrees of freedom. Changing the hypothesized mean difference changes the numerator of the test statistic, not the data or pooled variance.

本题组中：均值或均值差常选t；两个正态总体方差的比值检验选F；一个正态总体的方差检验可选卡方。先看检验对象，不要只凭出现“两个总体”就选F。

In these questions, t is the usual choice for means or mean differences, F for a ratio of two normal-population variances, and chi-square for the variance of one normal population. Identify the tested parameter rather than choosing F merely because two populations are mentioned.

> t_paired = (d_bar - d0) / (s_d / sqrt(n)); df = n - 1
>
> d_bar为样本配对差的均值，s_d为差值的样本标准差。
>
> d_bar is the sample mean of paired differences and s_d is their sample standard deviation.

t_pooled = [(x_bar1 - x_bar2) - delta0] / [s_p x sqrt(1/n1 + 1/n2)]

delta0为假设的均值差；df = n1 + n2 - 2。

delta0 is the hypothesized mean difference; df = n1 + n2 - 2.

s_p^2 = [(n1 - 1)s1^2 + (n2 - 1)s2^2] / (n1 + n2 - 2)

合并方差由样本大小与样本方差决定，不由假设均值差决定。

The pooled variance depends on sample sizes and sample variances, not the hypothesized mean difference.

Q05 均值差的检验统计量 / Statistic for a mean-difference test

巩固练习 / Reinforcement • 原题 PDF p.115 / 章节 p.1 • 答案 PDF p.5

5. Which of the following test statistics is *most* appropriate for a hypothesis test concerning the mean difference between two normally distributed populations?
   A. t-statistic.
   B. *F*-statistic.
   C. Chi-square statistic.

中文题干与选项 / Chinese translation

对两个正态总体的均值差进行假设检验，以下哪一种统计量最合适?

A. t统计量。

B. F统计量。

C. 卡方统计量。

答案 / Answer: A

1. 本题要检验均值差，在所给选项中应选t统计量。

The tested quantity is a mean difference, making the t statistic the appropriate option here.

2. F常用于比较两个正态总体的方差；卡方可用于单个正态总体方差或分类变量独立性，因此B、C不匹配本题对象。

F is commonly used to compare two normal-population variances, while chi-square can test one normal-population variance or independence of categorical variables. Neither matches the requested mean difference.

易错点 / Common trap

均值差选t，但具体公式仍需看配对还是独立、以及方差假设。

Choose t for the mean difference, but determine the exact formula from pairing, independence, and variance assumptions.

资料提示 / Source note

原解析用配对样本公式作说明，题干本身未明确写出配对关系；不能仅凭这道题据此认定所有均值差检验都用n - 1自由度。

The supplied explanation illustrates a paired-sample formula, but the stem does not explicitly specify pairing. Do not infer that all mean-difference tests have n - 1 degrees of freedom.

Q34 配对t检验自由度 / Degrees of freedom for a paired t-test

巩固练习 / Reinforcement · 原题 PDF p.121 / 章节 p.7 · 答案 PDF p.34

34. An analyst examines 30 paired monthly returns for two stock indexes. To determine if the mean difference of the returns is zero, the number of degrees of freedom of the t-test is.
    A. 28.
    B. 29.
    C. 58.

中文题干与选项 / Chinese translation

分析师研究两个股票指数30组配对月收益率。为了检验平均收益差是否为零，t检验的自由度为：

A. 28。

B. 29。

C. 58。

答案 / Answer: B

1. 每个月两个指数构成一对，先求30个配对差值。

The two index returns in each month form one pair, producing 30 paired differences.

2. 对这30个差值做均值检验：df = n - 1 = 30 - 1 = 29。28属于n - 2；58把两组错误当作独立样本。

Test the mean of these 30 differences: df = n - 1 = 30 - 1 = 29. Using 28 confuses this with n - 2; 58 incorrectly treats the two samples as independent.

易错点 / Common trap

n是配对数量30，不是把两个指数的观测数相加为60。

n is the number of pairs, 30, not the combined count of 60 index observations.

Q82 改变假设均值差 / Changing the hypothesized mean difference

巩固练习 / Reinforcement · 原题 PDF p.131 / 章节 p.17 · 答案 PDF p.82

82. An analyst performs a hypothesis test concerning the difference between the mean returns of two portfolios, assuming normally distributed populations with unknown but equal variances. If the analyst decides to change the hypothesized difference in mean returns from 0% to 1%, which of the following will change?
    A. The value of the test statistic.
    B. The degrees of freedom used in the test.
    C. The pooled estimate of the common population variance.

中文题干与选项 / Chinese translation

分析师检验两个组合平均收益的差异，假设两个总体正态、方差未知但相等。如果把假设的平均收益差从0%改为1%，以下哪项会改变？

A. 检验统计量的数值。

B. 检验使用的自由度。

C. 总体共同方差的合并估计值。

答案 / Answer: A

1. 合并方差t统计量的分子为(x_bar1 - x_bar2) - delta0；delta0从0%变为1%直接改变分子。

The numerator of the pooled-variance t statistic is (x_bar1 - x_bar2) - delta0. Changing delta0 from 0% to 1% directly changes that numerator.

2. 样本量和样本方差均没有改变，所以df = n1 + n2 - 2和合并方差s_p^2保持不变。

The sample sizes and sample variances do not change, so df = n1 + n2 - 2 and the pooled variance s_p^2 remain unchanged.

易错点 / Common trap

改变假设值不是改变数据；不要重新改动样本方差或自由度。

Changing a hypothesized value does not change the observations, sample variances, or degrees of freedom.

7.3 相关性与独立性：三种工具别混用 / Correlation and independence: three distinct tools

先学知识点 / Learn the concept first

Pearson相关系数检验在线性相关为零的原假设下，可使用t统计量，自由度为n - 2；常规精确推断要求相应的正态与独立抽样条件。

A test of zero Pearson correlation can use a t statistic with n - 2 degrees of freedom. Conventional exact inference requires the relevant normality and independent-sampling assumptions.

Spearman使用排名而非原始数值。没有并列排名时，可以用每一对排名差的平方和直接计算。

Spearman correlation uses ranks rather than raw values. When there are no tied ranks, it can be calculated directly from the sum of squared rank differences.

列联表检验两个分类变量是否独立，使用卡方统计量；自由度是(行数 - 1)(列数 - 1)。

A contingency-table test examines whether two categorical variables are independent using a chi-square statistic. Its degrees of freedom are (rows - 1)(columns - 1).

> t = r x sqrt(n - 2) / sqrt(1 - r^2); H0: rho = 0
>
> 分母是sqrt(1 - r^2)，不要漏掉平方。
>
> The denominator is sqrt(1 - r^2); do not omit the square.

> r_s = 1 - 6 sum(d_i^2) / [n(n^2 - 1)]
>
> d_i为同一观测的两项排名之差；此简式适用于没有并列排名。
>
> d_i is the difference between the two ranks for an observation; this shortcut assumes no tied ranks.

> df_chi_square = (rows - 1)(columns - 1)
>
> 不是行列数相乘，也不是行列数之和减二。
>
> It is neither rows times columns nor rows plus columns minus two.

Q17 零相关系数的t检验 / t-test of zero correlation

核心例题 / Core example · 原题 PDF p.118 / 章节 p.4 · 答案 PDF p.17

17. In a parametric test of the correlation between two variables with a sample size of 51 and sample correlation of 0.6, the t-statistic is *closest* to:
    A. 0.07.
    B. 5.25.
    C. 6.64.

中文题干与选项 / Chinese translation

对两个变量的相关性进行参数检验。样本量为51，样本相关系数为0.6，t统计量最接近：

A. 0.07。

B. 5.25。

C. 6.64。

答案 / Answer: B

1. 检验H0: rho = 0时，t = r sqrt(n - 2) / sqrt(1 - r^2)。

For H0: rho = 0, t = r sqrt(n - 2) / sqrt(1 - r^2).

2. 代入：sqrt(51 - 2) = 7，sqrt(1 - 0.6^2) = 0.8；所以t = 0.6 × 7 / 0.8 = 5.25。

Substitute sqrt(51 - 2) = 7 and sqrt(1 - 0.6^2) = 0.8; therefore t = 0.6 × 7 / 0.8 = 5.25.

易错点 / Common trap

C的6.64来自把分母错写成sqrt(1 - r)，遗漏r的平方。

C’ s 6.64 results from incorrectly using sqrt(1 - r), omitting the square on r.

Q83 Spearman等级相关 / Spearman rank correlation

核心例题 / Core example • 原题 PDF p.131 / 章节 p.17 • 答案 PDF p.83

83. An analyst tabulates the ranks of four paired observations of random variables X and Y as follows:

Observation	Rank of X	Rank of Y
1	1	2
2	2	3
3	3	4
4	4	1

The Spearman rank correlation coefficient between X and Y is *closest* to:

A. -0.2.

B. 0.8.

C. 1.0.

中文题干与选项 / Chinese translation

分析师把随机变量X与Y的四组配对观测排名如下。X与Y的Spearman等级相关系数最接近：

观测	X的排名	Y的排名
1	1	2
2	2	3
3	3	4
4	4	1

A. -0.2。

B. 0.8。

C. 1.0。

答案 / Answer: A

1. 四个排名差X - Y依次为-1、-1、-1、3，因此平方和为1 + 1 + 1 + 9 = 12。

The rank differences X - Y are -1, -1, -1, and 3, giving a squared-difference sum of 1 + 1 + 1 + 9 = 12.

2. 没有并列排名，$r_s = 1 - 6 \times 12 / [4 \times (4^2 - 1)] = 1 - 72/60 = -0.2$。

With no tied ranks, $r_s = 1 - 6 \times 12 / [4 \times (4^2 - 1)] = 1 - 72/60 = -0.2$.

易错点 / Common trap

排名差必须先逐项平方再求和；先相加会得到0并误判为完全相关。

Square each rank difference before summing. Adding the differences first gives zero and falsely suggests perfect correlation.

Q16 卡方独立性检验自由度 / Degrees of freedom for a chi-square independence test

巩固练习 / Reinforcement • 原题 PDF p.117 / 章节 p.3 • 答案 PDF p.16

16. A test of independence is based on the data in a contingency table with 5 rows and 4 columns. Using a nonparametric test statistic that is chi-square distributed, the number of degrees of freedom is:
    A. 7.
    B. 12.
    C. 20.

中文题干与选项 / Chinese translation

独立性检验使用一张5行4列的列联表。采用服从卡方分布的非参数检验统计量时，自由度是多少?

A. 7。

B. 12。

C. 20。

答案 / Answer: B

1. 公式：df = (r - 1)(c - 1)。

Formula: df = (r - 1)(c - 1).

2. 代入r = 5、c = 4：df = 4 × 3 = 12。A把行列减一后相加，C直接把行列相乘，均不正确。

With r = 5 and c = 4, df = 4 × 3 = 12. A adds the reduced row and column counts, while C multiplies the unreduced counts; both are incorrect.

易错点 / Common trap

这是列联表的自由度，不是两个独立样本均值差的n1 + n2 - 2。

These are contingency-table degrees of freedom, not n1 + n2 - 2 for an independent-sample mean-difference test.

7.4 何时考虑非参数检验 / When to consider nonparametric tests

先学知识点 / Learn the concept first

排名或顺序尺度、明显离群值、难以满足所需分布假设，或检验对象不是某个参数时，非参数方法通常更合适。

Nonparametric methods are often useful for ranks or ordinal data, influential outliers, failure of required distributional assumptions, or hypotheses that do not concern a specific parameter.

“数据不正态”不自动意味着必须用非参数方法。对均值推断，足够大的样本可能允许近似参数检验；但足够大取决于偏度和尾部，50不是适用于所有分布的保证。

Non-normal data do not automatically require a nonparametric method. For mean inference, sufficiently large samples may allow approximate parametric tests, but the required size depends on skewness and tails; 50 is not a universal guarantee.

Q90 排名数据与非参数检验 / Rank data and nonparametric tests

巩固练习 / Reinforcement • 原题 PDF p.132 / 章节 p.18 • 答案 PDF p.90

90. A nonparametric test is *most* appropriate when:
    A. comparing differences between means.
    B. data are given in ranks.
    C. data meet distributional assumptions.

中文题干与选项 / Chinese translation

以下哪种情形最适合使用非参数检验?

A. 比较均值之间的差。

B. 数据以排名形式给出。

C. 数据满足分布假设。

答案 / Answer: B

1. 排名只提供顺序信息，适合使用基于排名的非参数方法。

Ranks provide ordering information and are well suited to rank-based nonparametric methods.

2. 均值差本身并不要求非参数方法；如果所需分布假设已经满足，参数检验通常可以使用。

A mean difference does not by itself require a nonparametric method. If the required distributional assumptions are satisfied, a parametric test is generally available.

易错点 / Common trap

nonparametric并不是“不计算统计量”，而是对特定参数或分布形式的依赖较少。

Nonparametric does not mean no test statistic; it means less dependence on a particular parameterization or distributional form.

Q04 参数检验与非参数检验 / Parametric versus nonparametric tests

巩固练习 / Reinforcement • 原题 PDF p.115 / 章节 p.1 • 答案 PDF p.4

4. For a sample of 50 observations, in which of the following situations is a nonparametric test least *likely* to be appropriate? The data:
    A. contain outliers.
    B. are given in ranks.
    C. come from a population with a log normal distribution.

中文题干与选项 / Chinese translation

对于含50个观测值的样本，在以下哪种情况下，非参数检验最不可能合适？数据：

A. 包含离群值。

B. 以排名形式给出。

C. 来自服从对数正态分布的总体。

答案 / Answer: C

1. A的离群值和B的排名都属于使用非参数方法的常见理由。

Outliers in A and rank data in B are common reasons to use nonparametric methods.

2. 原答案按50个观测属于较大样本的考试口径，认为可以对均值采用近似参数检验，因此在三个选项中选C。

The supplied solution treats 50 observations as a relatively large sample for an approximate parametric mean test, making C the intended choice among the three options.

易错点 / Common trap

题目问least likely；要选最弱的非参数检验理由。

The question asks least likely; select the weakest reason for choosing a nonparametric test.

资料提示 / Source note

这不是“50个样本一定足够”的定理。极强偏度或厚尾时，样本均值的正态近似仍可能较差；实际使用需检查分布与检验条件。

This is not a theorem that 50 observations are always sufficient. With severe skewness or heavy tails, the normal approximation for the mean may still be poor, so practical use requires checking the distribution and test conditions.

单元 08 简单线性回归

Simple Linear Regression

把回归题拆成五步：谁解释谁、是否线性、误差多大、关系是否显著、如何预测。先认变量和单位，再计算。

Break regression questions into five steps: which variable explains which, whether the relationship is linear, how large the errors are, whether the relationship is significant, and how to predict. Identify the variables and units before calculating.

本单元 13 题 / 13 questions in this unit

单元速记 / Unit essentials

Y是被解释变量；残差等于实际Y减拟合Y。

Y is the explained variable; a residual is actual Y minus fitted Y.

SEE使用SSE和n - 2；R^2使用解释变动占总变动的比例。

SEE uses SSE and n - 2; R-squared is the explained share of total variation.

预测区间以预测Y为中心；显著性水平上升意味着置信水平下降。

Center a prediction interval on predicted Y; increasing significance lowers confidence.

8.1 模型、残差和基本假设 / Model, residuals, and core assumptions

先学知识点 / Learn the concept first

Y是因变量、被解释变量或响应变量；X是自变量或解释变量。模型用X预测Y。残差是观测Y减去样本拟合值。

Y is the dependent, explained, or response variable. X is the independent or explanatory variable. The model predicts Y from X. A residual is observed Y minus the sample-fitted value.

线性模型适当时，残差对X的图应在零附近随机散布，没有系统性曲线。这个图也用于检查方差是否大致稳定。

With an appropriate linear model, a plot of residuals against X should show random scatter around zero without a systematic curve. It also helps check whether residual variance is roughly constant.

常规简单回归假设包括线性条件均值、零条件均值误差、同方差和观测间不相关的误差。正态误差支持常规小样本t和F精确推断；仅计算OLS拟合线不必要求Y边际正态。

The conventional simple-regression model assumes a linear conditional mean, errors with zero conditional mean, constant error variance, and uncorrelated errors across observations. Normal errors support conventional exact small-sample t and F inference; merely fitting an OLS line does not require marginal normality of Y.

$$Y_i = b0 + b1\ X_i + error_i;\ fitted\ Y_i = b0_hat + b1_hat\ X_i$$

b0是截距，b1是斜率；估计值与总体参数要区分。

b0 is the intercept and b1 is the slope; distinguish estimates from population parameters.

$$residual_i = Y_i - fitted\ Y_i$$

残差可由样本计算；总体真实误差通常不可观察。

Residuals can be computed from the sample; the true population errors are generally unobservable.

Q84 被解释变量的识别 / Identifying the explained variable

巩固练习 / Reinforcement • 原题 PDF p.131 / 章节 p.17 • 答案 PDF p.84

84. An analyst runs a simple linear regression to test whether the variation in the demand for corn explains the variation in the supply of wheat. In this model, the supply of wheat is a(n):
    A. indicator variable.
    B. explained variable.
    C. independent variable.

中文题干与选项 / Chinese translation

分析师进行简单线性回归，检验玉米需求的变动能否解释小麦供给的变动。在该模型中，小麦供给是：

A. 指示变量。

B. 被解释变量。

C. 自变量。

答案 / Answer: B

1. 句子结构是“玉米需求解释小麦供给”，因此玉米需求是X，小麦供给是Y。

The wording says corn demand explains wheat supply, so corn demand is X and wheat supply is Y.

2. Y又叫因变量、响应变量或被解释变量，故选B。

Y is also called the dependent, response, or explained variable, so B is correct.

易错点 / Common trap

不要因为经济直觉认为供给应影响需求，就反转题目明确指定的回归方向。

Do not reverse the explicitly stated regression direction based on an economic intuition that supply might affect demand.

Q68 残差与总体误差 / Residual versus population error

巩固练习 / Reinforcement • 原题 PDF p.128 / 章节 p.14 • 答案 PDF p.68

68. With respect to simple linear regression, a residual is best described as the difference between the observed value of a dependent variable and:
    A. its mean.
    B. its estimated value using a fitted regression line based on the sample.
    C. its expected value based on the true underlying population relationship.

中文题干与选项 / Chinese translation

简单线性回归中的残差，最恰当地描述为因变量观测值减去：

A. 该因变量的均值。

B. 根据样本拟合回归线得到的估计值。

C. 根据真实总体关系得到的期望值。

答案 / Answer: B

1. 残差e_i = Y_i - fitted Y_i，由实际样本和估计回归线计算。

The residual e_i = Y_i - fitted Y_i is calculated from the observed sample and estimated regression line.

2. A是相对样本均值的总偏差；C对应不可直接观察的总体真实误差，不是样本残差。

A is the total deviation from the sample mean. C corresponds to the generally unobservable true population error, not the sample residual.

易错点 / Common trap

residual是样本拟合误差；error term指总体模型误差。

A residual is the sample fitting error; an error term belongs to the population model.

Q86 残差图诊断 / Residual-plot diagnostics

巩固练习 / Reinforcement • 原题 PDF p.132 / 章节 p.18 • 答案 PDF p.86

86. If the relationship between the dependent variable and independent variable is linear, the regression residuals when plotted against the independent value should appear to:
    A. be linear.
    B. be random.
    C. follow a pattern.

中文题干与选项 / Chinese translation

如果因变量与自变量之间的关系是线性的，把回归残差对自变量绘图，残差应呈现：

A. 线性排列。

B. 随机分布。

C. 某种规律性模式。

答案 / Answer: B

1. 拟合线已经吸收线性关系，剩余残差应在零附近无系统模式地散布。

The fitted line has absorbed the linear relationship, leaving residuals scattered around zero without a systematic pattern.

2. 若残差图仍出现弯曲或其他规律，说明模型可能遗漏结构；残差不应再排成一条有趋势的线。

A curved or other systematic residual pattern suggests that the model may have missed structure. Residuals should not still form a trending line.

易错点 / Common trap

原始(X,Y)散点可能沿直线排列，但残差图应随机散布。

The original (X,Y) points may align along a line; the residual plot should show random scatter.

Q80 回归的常规分布假设 / Conventional distributional assumptions in regression

巩固练习 / Reinforcement • 原题 PDF p.131 / 章节 p.17 • 答案 PDF p.80

80. Which of the following is an underlying assumption of the simple linear regression model? The regression residuals:
    A. are normally distributed.
    B. have high correlations across observations.
    C. have different variances across observations.

中文题干与选项 / Chinese translation

以下哪项是简单线性回归模型的一项基本假设？回归残差：

A. 服从正态分布。

B. 在不同观测之间具有很高的相关性。

C. 在不同观测之间具有不同的方差。

答案 / Answer: A

1. 按题库的常规简单回归推断框架，答案为A。B违反误差不相关假设，C违反同方差假设。

Under the question bank’s conventional simple-regression inference framework, the answer is A. B violates uncorrelated errors and C violates constant error variance.

2. 更准确地说，模型对总体误差项作条件正态性假设，用样本残差来诊断；正态性用于常规小样本推断，不是算出OLS系数的必要条件。

More precisely, the model assumes conditional normality of the population errors and uses sample residuals for diagnostics. Normality supports conventional small-sample inference; it is not required merely to compute OLS coefficients.

易错点 / Common trap

不要把题目的简化表述扩展成“所有回归都要求原始Y正态”。

Do not generalize this simplified wording into a claim that every regression requires raw Y to be normally distributed.

资料提示 / Source note

原题使用residuals一词。本讲义保留题干与A答案，并区分理论误差项与估计后的样本残差。

The original stem uses the word residuals. The stem and answer A are retained while distinguishing theoretical errors from estimated sample residuals.

8.2 非线性关系与对数变换 / Nonlinear relationships and logarithmic transformations

先学知识点 / Learn the concept first

关系有曲线时，可以考虑变换X、Y或两者，让变换后的关系适合线性回归；不是只要Y不正态就必须取对数。

If the relationship is curved, transforming X, Y, or both may make the transformed relationship suitable for linear regression. A non-normal marginal distribution of Y alone does not automatically require a logarithmic transformation.

命名顺序先Y后X：lin-log只对X取对数；log-lin只对Y取对数；log-log对两者取对数。

Names list Y first and X second: lin-log logs only X, log-lin logs only Y, and log-log logs both.

Lin-log: $Y = b0 + b1 \ln(X) + error$

仅自变量为对数形式。

Only the independent variable is logged.

Log-lin: $\ln(Y) = b0 + b1 X + error$

仅因变量为对数形式。

Only the dependent variable is logged.

Log-log: $\ln(Y) = b0 + b1 \ln(X) + error$

两边都取对数。

Both variables are logged.

Q20 线性化变换 / Transformations to achieve linearity

巩固练习 / Reinforcement • 原题 PDF p.118 / 章节 p.4 • 答案 PDF p.20

20. Which of the following best describes when a transformation of the data may be needed to enable the use of a simple linear regression model? When the:
    A. dependent variable is non-normally distributed.
    B. pairs of the dependent and independent variables are uncorrelated with one another.
    C. relationship between the independent variable and the dependent variable is non-linear.

中文题干与选项 / Chinese translation

在以下哪种情况下，最可能需要对数据作变换，才能使用简单线性回归模型？

A. 因变量不服从正态分布。

B. 不同的因变量与自变量配对观测之间不相关。

C. 自变量与因变量之间的关系是非线性的。

答案 / Answer: C

1. 简单线性回归要求模型中的关系能用线性形式表示；明显的曲线关系可能通过适当变换线性化。

Simple linear regression requires a relationship that can be represented in linear form within the model. A curved relationship may become linear after a suitable transformation.

2. Y的边际分布不正态并不自动破坏线性条件均值模型；B说的是不同(X,Y)观测对之间不相关，并不是说X与Y本身不相关，这也不构成必须变换的理由。

A non-normal marginal distribution of Y does not automatically invalidate a linear conditional-mean model. B refers to different (X,Y) observation pairs being uncorrelated with one another, not X being uncorrelated with Y; this is not a reason that a transformation is required.

易错点 / Common trap

检查的是变量之间的关系是否线性，而不是只检查Y的直方图是否正态。

Check whether the relationship between the variables is linear, rather than only whether Y’ s histogram is normal.

资料提示 / Source note

原解析用“不相关”直接推出“独立”，一般并不成立；独立通常能推出不相关（相关矩存在时），反向需要额外条件。此处不影响C答案。

The supplied explanation directly infers independence from zero correlation, which is not generally valid. Independence implies zero correlation when the relevant moments exist, but the converse needs additional conditions. Answer C is unaffected.

Q56 对数模型命名 / Naming logarithmic models

巩固练习 / Reinforcement • 原题 PDF p.126 / 章节 p.12 • 答案 PDF p.56

56. The simple linear regression model in which only the independent variable is in logarithmic form is *best* described as the.
    A. log-lin model.
    B. lin-log model.
    C. log-log model

中文题干与选项 / Chinese translation

在一个简单线性回归模型中，只有自变量采用对数形式，这个模型称为：

A. log-lin模型。

B. lin-log模型。

C. log-log模型。

答案 / Answer: B

1. 模型形式为Y = b0 + b1 ln(X) + error。

The model is Y = b0 + b1 ln(X) + error.

2. 名称先写Y再写X：Y保持线性形式lin，X取对数log，所以是lin-log。

The name lists Y before X: Y remains in levels, lin, while X is logged, log; hence lin-log.

易错点 / Common trap

不要按“先想到自变量”来命名；固定按因变量Y、自变量X的顺序。

Do not name the model by whichever variable comes to mind first; consistently use dependent Y, then independent X.

8.3 图形、平方和与估计标准误 / Plots, sums of squares, and standard error of the estimate

先学知识点 / Learn the concept first

散点图直接展示两变量的配对观测，可判断方向、线性程度和离群值。

A scatter plot displays paired observations of two variables, revealing direction, linearity, and outliers.

含截距的OLS简单回归中：总变动SST = 解释变动SSR + 未解释变动SSE。SST衡量Y围绕其样本均值的变动。

For OLS simple regression with an intercept, total variation SST equals explained variation SSR plus unexplained variation SSE. SST measures variation of Y around its sample mean.

估计标准误SEE衡量观测Y到拟合线的典型距离，单位与Y相同；R^2才是被解释变动所占比例。

The standard error of the estimate measures the typical distance from observed Y to the fitted line and has the same units as Y. R-squared, instead, measures the proportion of variation explained.

SST = sum((Y_i - Y_bar)^2) = SSR + SSE

本讲义SSR指回归解释平方和，SSE指误差平方和。

Here SSR denotes the regression sum of squares and SSE the error sum of squares.

SEE = sqrt[SSE / (n - 2)]

估计了截距和斜率两个参数，所以误差自由度为n - 2。

Estimating an intercept and a slope uses two parameters, leaving n - 2 residual degrees of freedom.

R^2 = SSR / SST = 1 - SSE / SST

R^2没有量纲，不能与SEE混淆。

R-squared is unitless and should not be confused with SEE.

Q30 相关性的图形检查 / Graphical inspection of correlation

巩固练习 / Reinforcement • 原题 PDF p.120 / 章节 p.6 • 答案 PDF p.30

30. Which of the following visualizations is *most* appropriate for interpreting the correlation between two variables?
    A. Tree-map.
    B. Scatter plot.
    C. Clustered bar chart.

中文题干与选项 / Chinese translation

以下哪种可视化最适合解读两个变量之间的相关性？

A. 矩形树图。

B. 散点图。

C. 簇状条形图。

答案 / Answer: B

1. 把每组配对观测画成(X,Y)一个点，散点图可以直接显示变量共同变化的方向与形态。

Plot each paired observation as a point (X,Y); a scatter plot directly shows the direction and shape of joint variation.

2. 矩形树图侧重构成与层级，簇状条形图侧重类别比较，都不如散点图适合观察连续变量的相关关系。

Tree-maps emphasize composition and hierarchy, while clustered bars compare categories. Neither is as suitable as a scatter plot for examining correlation between continuous variables.

易错点 / Common trap

相关性强不等于因果关系成立。

Strong correlation does not establish causation.

Q76 总平方和 / Total sum of squares

巩固练习 / Reinforcement • 原题 PDF p.130 / 章节 p.16 • 答案 PDF p.76

76. In simple linear regression analysis, the total sum of squares best describes:
    A. a scatter plot.
    B. the variation of the dependent variable.
    C. a paired observation between variables.

中文题干与选项 / Chinese translation

简单线性回归分析中的总平方和，最恰当地描述了：

A. 一张散点图。

B. 因变量的变动。

C. 变量之间的一组配对观测。

答案 / Answer: B

1. SST = sum[(Y_i - Y_bar)^2]，把所有Y相对均值的偏差平方后相加。

SST = sum[(Y_i - Y_bar)^2], the sum of squared deviations of Y from its mean.

2. 因此它衡量因变量的总变动，可分解为模型解释部分与残差部分；它不是图形或单个数据点。

It therefore measures total variation in the dependent variable, which can be split into explained and residual components. It is neither a graph nor a single observation.

易错点 / Common trap

SST看Y绕均值的变动，SSE看Y绕拟合线的变动。

SST measures Y’s variation around its mean; SSE measures Y’s variation around the fitted line.

Q44 估计标准误的含义 / Interpreting the standard error of the estimate

巩固练习 / Reinforcement • 原题 PDF p.123 / 章节 p.9 • 答案 PDF p.44

44. The standard error of the estimate in a simple linear regression is *best* described as:
    A. a relative measure off it for the regression.
    B. the percentage of the variation of the dependent variable that is explained by the independent variable.
    C. a measure of the distance between the observed values of the dependent variable and those predicted from the estimated regression.

中文题干与选项 / Chinese translation

简单线性回归的估计标准误，最恰当地描述为：

A. 回归拟合程度的相对度量。

B. 自变量所解释的因变量变动百分比。

C. 因变量观测值与估计回归预测值之间距离的度量。

答案 / Answer: C

1. SEE来自残差平方和，衡量实际Y与拟合Y之间典型偏离的大小，因此C正确。

SEE is based on the residual sum of squares and measures the typical size of deviations between actual and fitted Y, making C correct.

2. SEE与Y的单位相同，是绝对误差尺度；B描述的是R^2，属于相对拟合度量。

SEE has the same units as Y and is an absolute error measure. B describes R-squared, a relative goodness-of-fit measure.

易错点 / Common trap

SEE不是百分比解释力；不要与R^2混用。

SEE is not the percentage of variation explained; do not confuse it with R-squared.

Q23 估计标准误计算 / Calculating the standard error of the estimate

核心例题 / Core example • 原题 PDF p.119 / 章节 p.5 • 答案 PDF p.23

23. An analyst estimates the following information from a simple linear regression:

Sum of squares error	280
Sum of squares regression	25
Number of paired observations	30

The standard error of the estimate is *closest* to:

A. 2.5.

B. 3.2.

C. 10.0.

中文题干与选项 / Chinese translation

分析师从简单线性回归得到：误差平方和280、回归平方和25、配对观测数30。估计标准误最接近：

A. 2.5。

B. 3.2。

C. 10.0。

答案 / Answer: B

1. 使用SEE = sqrt[SSE/(n - 2)]；本题SSE = 280、n = 30。

Use SEE = sqrt[SSE/(n - 2)], with SSE = 280 and n = 30.

2. SEE = sqrt(280/28) = sqrt(10) = 3.1623，最接近3.2。回归解释平方和25不进入这一公式。

SEE = sqrt(280/28) = sqrt(10) = 3.1623, closest to 3.2. The regression sum of squares of 25 is not used in this formula.

易错点 / Common trap

10.0是残差均方MSE，最后还必须开平方得到SEE。

10.0 is the residual mean square, MSE; take its square root to obtain SEE.

8.4 回归显著性：F检验的是斜率 / Regression significance: the F-test concerns the slope

先学知识点 / Learn the concept first

简单线性回归只有一个解释变量，整体F检验的原假设是斜率为零，表示X没有线性解释力；不是检验截距为零。

With one explanatory variable, the overall F-test in simple regression tests a zero slope, meaning no linear explanatory power of X. It is not a test of a zero intercept.

在同一模型与同一原假设下，简单回归整体F统计量等于斜率双尾t检验统计量的平方。

For the same model and null hypothesis, the overall F statistic in simple regression equals the squared t statistic for the two-sided slope test.

H0: slope = 0; F = (SSR / 1) / [SSE / (n - 2)] = t_slope^2

F检验自由度为(1, n - 2)。

The F-test has (1, n - 2) degrees of freedom.

Q27 简单回归的F检验 / F-test in simple regression

巩固练习 / Reinforcement • 原题 PDF p.120 / 章节 p.6 • 答案 PDF p.27

27. The null hypothesis for the F-distributed test statistic in a simple linear regression model tests whether the:
    A. slope is equal to zero.
    B. intercept is equal to zero.
    C. slope is not equal to zero.

中文题干与选项 / Chinese translation

简单线性回归模型中，服从F分布的检验统计量所检验的原假设是：

A. 斜率等于零。

B. 截距等于零。

C. 斜率不等于零。

答案 / Answer: A

1. 只有一个解释变量时，整体回归F检验的H0为总体斜率等于零。

With one explanatory variable, the overall regression F-test has a null hypothesis of zero population slope.

2. 斜率不等于零是备择假设；截距是否为零不是该整体F检验的对象。

A nonzero slope is the alternative hypothesis. Whether the intercept is zero is not the object of this overall F-test.

易错点 / Common trap

F值越大通常说明反对零斜率的证据越强；但H0本身仍是“等于零”。

A larger F value generally provides stronger evidence against zero slope, but the null itself is still equality to zero.

8.5 预测值与预测区间 / Predicted values and prediction intervals

先学知识点 / Learn the concept first

先把预测的X代入回归线得到预测Y，再围绕预测Y加减临界值乘预测标准误。题目给了预测标准误时直接使用，不要换成X或遗漏t临界值。

First substitute the forecast X into the fitted line to obtain predicted Y. Then add and subtract the critical value times the forecast standard error around predicted Y. If the forecast standard error is given, use it directly; do not center the interval on X or omit the t critical value.

其他条件相同，估计标准误越大，预测区间越宽。更高置信水平（更低显著性水平）通常也使区间变宽；更多样本通常改善精度。

Other things equal, a larger estimation standard error widens the prediction interval. Higher confidence, which means lower significance, also usually widens it; more observations generally improve precision.

预测单个未来观测的区间包含未来随机误差，一般比相同X处平均响应的置信区间更宽。

A prediction interval for one future observation includes future random error and is generally wider than a confidence interval for the mean response at the same X.

$$Y_{forecast} = b0_hat + b1_hat\ X_{forecast}$$

先算因变量预测值。

Calculate the predicted dependent variable first.

Prediction interval = Y_forecast +/- t_critical x s_forecast

s_forecast是预测标准误；不是把SEE和预测标准误视为同一个量。

s_forecast is the forecast standard error; SEE and forecast standard error are distinct quantities.

Q02 回归预测区间 / Regression prediction interval

核心例题 / Core example · 原题 PDF p.115 / 章节 p.1 · 答案 PDF p.2

2. An analyst performs a simple linear regression of a stock's monthly return on the monthly return of a market index (both in %) and gathers the following information:

Estimated slope	1.0
Estimated intercept	1.2%
Standard error of the forecast	1.4%
Critical t-values at a 5% significance level	±2.032

The 95% prediction interval for the stock's monthly return, given that the forecasted monthly return on the index is 3.5%, is *closest* to:

A. 0.7% to 6.3%.

B. 1.9% to 7.5%.

C. 3.3% to 6.1%.

中文题干与选项 / Chinese translation

分析师以市场指数月收益率解释某股票月收益率（两者均以%表示），估计斜率为1.0、截距为1.2%、预测标准误为1.4%，5%显著性水平下的t临界值为±2.032。如果预测指数月收益率为3.5%，该股票月收益率的95%预测区间最接近：

A. 0.7%至6.3%。

B. 1.9%至7.5%。

C. 3.3%至6.1%。

答案 / Answer: B

1. 先算股票预测收益：Y_f = 1.2% + 1.0 × 3.5% = 4.7%。

First calculate the predicted stock return: Y_f = 1.2% + 1.0 × 3.5% = 4.7%.

2. 误差界限 = 2.032 × 1.4% = 2.8448个百分点；区间 = 4.7% ± 2.8448% = [1.8552%, 7.5448%]，约为[1.9%, 7.5%]。

The margin is 2.032 × 1.4% = 2.8448 percentage points. The interval is 4.7% ± 2.8448% = [1.8552%, 7.5448%], approximately [1.9%, 7.5%].

3. A错误地以指数预测收益3.5%为中心；C遗漏t临界值，只加减一次预测标准误。

A incorrectly centers the interval on the predicted index return of 3.5%. C omits the t critical value and adds and subtracts only one forecast standard error.

易错点 / Common trap

先求Y，再加减t × s_f；不要直接围绕X构造区间。

Find Y first, then add and subtract t × s_f; do not build the interval directly around X.

Q65 预测区间宽度 / Width of a prediction interval

巩固练习 / Reinforcement • 原题 PDF p.128 / 章节 p.14 • 答案 PDF p.65

65. All else being equal, which of the following would *most likely* lead to a wider prediction interval for the dependent variable when re-estimating a linear regression model? An increase in the:
    A. sample size.
    B. level of significance.
    C. standard error of the estimate.

中文题干与选项 / Chinese translation

其他条件相同，重新估计线性回归模型时，以下哪项增加最可能使因变量的预测区间变宽?

A. 样本量。

B. 显著性水平。

C. 估计标准误。

答案 / Answer: C

1. 预测区间的误差界限与预测标准误相关，而预测标准误随着估计标准误SEE增加而增加。

The prediction interval’s margin depends on the forecast standard error, which increases with the estimation standard error, SEE.

2. 样本量增加通常提高精度；显著性水平alpha提高意味着置信水平降低、临界值减小，所以都不是本题的变宽原因。

Increasing sample size generally improves precision. Increasing significance alpha lowers confidence and reduces the critical value. Neither is the widening factor sought here.

易错点 / Common trap

置信水平与显著性水平方向相反：confidence = 1 - alpha。

Confidence and significance move in opposite directions: confidence = 1 - alpha.

单元 09 金融科技与数据应用

Fintech and Data Applications

这一单元以识别概念为主。先把广义金融科技、数据分析工具、交易应用和分布式账本分开，再用题干中的任务词匹配相应技术。本单元保留原题库中的全部相关题。

This unit mainly tests concept recognition. Distinguish the broad fintech category, data-analysis tools, trading applications, and distributed ledgers, then match the task described in the question to the appropriate technology. All relevant items from the supplied question section are retained.

本单元 8 题 / 8 questions in this unit

单元速记 / Unit essentials

情绪识别选NLP；数据清理选data curation；资产权利上链选tokenization；实时自动交易关注low latency。

Match sentiment detection to NLP, data cleaning to data curation, asset rights on a ledger to tokenization, and real-time automated trading to low latency.

“仅过拟合才会错”“所有加密货币都无限发行”等绝对说法，是本组题的常见干扰项。

Absolute claims such as "only overfitting causes failure" or "all cryptocurrencies have unlimited supply" are common distractors in this set.

9.1 金融科技的范围与财富管理应用 / The scope of fintech and wealth-management applications

先学知识点 / Learn the concept first

Fintech是金融产品与服务在设计和交付方面的技术创新；大数据或算法交易只是其中的部分应用。

Fintech means technological innovation in designing and delivering financial products and services. Big Data and algorithmic trading are narrower applications within that broad area.

完全自动化财富管理适合标准化服务；复杂资产、特殊偏好及高度定制需求，是本题组中人类顾问的相对优势。

Fully automated wealth management is suited to standardized services. In these questions, complex assets, particular preferences, and extensive customization support the relative advantage of human advisers.

Q22 金融科技的定义 / Definition of fintech

巩固练习 / Reinforcement · 原题 PDF p.119 / 章节 p.5 · 答案 PDF p.22

22. In its broadest sense, fintech is *best* described as:
    A. the vast amount of data being generated by the financial services industry.
    B. the execution of investment strategies through computer-generated algorithms.
    C. technological innovation in the design and delivery of financial services and products.

中文题干与选项 / Chinese translation

从最广泛的意义看，金融科技最适合描述为：

A. 金融服务行业产生的海量数据。

B. 通过计算机生成的算法执行投资策略。

C. 金融产品与服务在设计及交付方面的技术创新。

答案 / Answer: C

1. 题干要求最广泛的定义，因此应选覆盖整个金融产品和服务创新的C。

The question asks for the broadest definition, so C correctly covers innovation across financial products and services.

2. A描述大数据，B描述自动化或算法交易；两者都是更狭窄的概念。

A describes Big Data and B describes automated or algorithmic trading; both are narrower concepts.

易错点 / Common trap

看到in its broadest sense，优先选择上位概念而不是单一应用。

The phrase in its broadest sense calls for the overarching category, not a single application.

Q14 传统顾问的相对优势 / Relative advantage of traditional financial advisers

巩固练习 / Reinforcement • 原题 PDF p.117 / 章节 p.3 • 答案 PDF p.14

14. Which of the following is *most likely* an advantage of traditional financial advisers over fully automated digital wealth managers?
    A. Lower account minimums.
    B. Dividend reinvestment options.
    C. Solutions that better address the needs of complex portfolios.

中文题干与选项 / Chinese translation

与完全自动化数字财富管理者相比，传统财务顾问最可能具有什么优势？

A. 更低的开户最低金额。

B. 提供股息再投资选项。

C. 提供更好满足复杂投资组合需求的方案。

答案 / Answer: C

1. 复杂组合可能涉及更多资产类型、特殊偏好与定制需求；原答案认为人类顾问在处理这些情况时有相对优势，选C。

Complex portfolios can involve more asset types, special preferences, and customization. The source identifies these needs as a relative advantage of human advisers, so C is correct.

2. 低开户门槛通常是自动化服务的特点；自动化服务也可提供股息再投资，因此A和B不是这里的相对优势。

Low account minimums are typically associated with automated services, and those services can also offer dividend reinvestment, so A and B do not identify the relative advantage here.

易错点 / Common trap

不要把普通自动化功能当作传统顾问独有的优势。

Do not treat ordinary automated functions as advantages unique to human advisers.

9.2 算法交易：速度与订单拆分 / Algorithmic trading: latency and order splitting

先学知识点 / Learn the concept first

低延迟表示数据传输和处理等待时间短。依赖实时价格的自动交易需要低延迟网络。

Low latency means little delay in transmitting and processing data. Automated trading that relies on real-time prices requires low-latency networks.

大额机构订单可被算法拆成多个小单，因此“机构大订单使用算法”不代表平均成交规模上升。

An algorithm may split a large institutional order into many small trades. The use of algorithms for large orders therefore does not imply an increase in average trade size.

Q11 算法交易与低延迟 / Algorithmic trading and low latency

巩固练习 / Reinforcement • 原题 PDF p.117 / 章节 p.3 • 答案 PDF p.11

11. Which of the following statements is *most* accurate with respect to the widespread adoption of algorithmic trading in financial markets?
    A. The need for low-latency networks has grown.
    B. Markets have become less fragmented in terms of trading venues.
    C. Average trade size has increased as algorithmic trading is used to execute large institutional orders.

中文题干与选项 / Chinese translation

关于金融市场广泛采用算法交易，下列哪项最准确?

A. 对低延迟网络的需求增加。

B. 从交易场所看，市场变得更不分散。

C. 由于算法交易用于执行机构大额订单，平均成交规模增加。

答案 / Answer: A

1. 基于实时价格和市场事件作出自动决策，需要尽量缩短数据传输和处理延迟，因此A正确。

Automated decisions based on real-time prices and market events require minimal transmission and processing delays, so A is correct.

2. 原答案指出交易场所趋于分散；算法还会拆分大额订单，所以大额机构订单的使用不意味着平均单笔成交更大。

The source notes fragmentation across trading venues and the splitting of large orders, so using algorithms for large institutional orders does not imply larger average trade sizes.

易错点 / Common trap

low latency是低延迟，不是低成交量；大订单可通过许多小单完成。

Low latency means little delay, not low trading volume; a large order can be executed through many small trades.

9.3 机器学习与文本情绪识别 / Machine learning and sentiment detection

先学知识点 / Learn the concept first

过度拟合把训练数据中的噪声也学进去，样本外预测变差；拟合不足未能学到数据中的重要关系。两者都可能导致预测失败。

Overfitting captures noise in the training data and harms out-of-sample prediction. Underfitting fails to capture important relationships. Either can cause inaccurate predictions.

识别评论中的情绪变化，用自然语言处理NLP；清理数据错误属于数据整理，不等同于情绪分析。

Use natural language processing, NLP, to detect changes in sentiment within commentary. Data curation cleans errors in data; it is not itself sentiment analysis.

Q12 过度拟合与拟合不足 / Overfitting and underfitting

核心例题 / Core example · 原题 PDF p.117 / 章节 p.3 · 答案 PDF p.12

12. The failure of machine learning models to accurately predict outcomes can be the result of:
    A. overfitting, but not underfitting.
    B. underfitting, but not overfitting.
    C. either overfitting or underfitting

中文题干与选项 / Chinese translation

机器学习模型不能准确预测结果，可能是由于：

A. 过度拟合，但不是拟合不足。

B. 拟合不足，但不是过度拟合。

C. 过度拟合或拟合不足中的任何一种。

答案 / Answer: C

1. 过度拟合时，模型过度贴合训练样本中的细节和噪声，在不同数据上预测失效。

With overfitting, the model follows details and noise in the training sample too closely and predicts poorly on different data.

2. 拟合不足时，模型没有充分识别真实数据关系；两种问题都可能导致预测不准，因此选C。

With underfitting, the model does not adequately capture the underlying relationships. Either problem can impair prediction, so C is correct.

易错点 / Common trap

训练样本拟合很好，不等于样本外预测好；模型过于简单也不保证可靠。

A good fit to training data does not guarantee good out-of-sample prediction, and an overly simple model is not necessarily reliable.

Q29 用自然语言处理识别情绪 / Using NLP to detect sentiment

巩固练习 / Reinforcement · 原题 PDF p.120 / 章节 p.6 · 答案 PDF p.29

29. Which of the following is *most likely* used to detect sentiment shifts in an analyst's commentary?
    A. Tokenization.
    B. Data curation.
    C. Natural language processing.

中文题干与选项 / Chinese translation

最可能用于识别分析师评论中情绪变化的是：

A. 代币化。

B. 数据整理。

C. 自然语言处理。

答案 / Answer: C

1. 评论是非结构化文本，识别其语气和情绪变化属于自然语言处理的应用，选C。

Commentary is unstructured text, and identifying shifts in tone and sentiment is an application of natural language processing, so C is correct.

2. 本题语境的tokenization指资产权利代币化；数据整理用于提高数据质量，本身并不完成情绪识别。

In the source context, tokenization refers to asset-rights tokenization. Data curation improves data quality but does not itself perform sentiment detection.

易错点 / Common trap

文本处理里也可能出现“分词”这个tokenization术语；本题问完整的情绪识别技术，仍选NLP。

Tokenization can also mean splitting text into tokens in language processing; the question asks for the broader sentiment-detection technology, which is NLP.

9.4 分布式账本、加密货币与代币化 / Distributed ledgers, cryptocurrencies, and tokenization

先学知识点 / Learn the concept first

题库所述加密货币可让交易双方在不依赖银行等中介的情况下交易；这不意味着其价格低波动，也不意味着所有加密货币都没有发行上限。

The cryptocurrencies described in the source can allow parties to transact without relying on an intermediary such as a bank. This does not imply low price volatility or an absence of supply limits for all cryptocurrencies.

本题组的资产代币化，是在分布式账本上表示实物资产所有权，可帮助转移该所有权。首次代币发行是融资活动；共识机制负责让网络对账本状态达成一致。

In these questions, asset tokenization represents ownership rights to physical assets on a distributed ledger and can facilitate ownership transfer. An initial coin offering is a fundraising activity; a consensus mechanism enables agreement about the ledger state.

Q07 加密货币的交易特征 / Transaction characteristics of cryptocurrencies

巩固练习 / Reinforcement • 原题 PDF p.116 / 章节 p.2 • 答案 PDF p.7

7. Which of the following statements is *most* accurate? Cryptocurrencies
   A. exhibit low volatility.
   B. have no limits on the total amount of currency that may be issued.
   C. allow transactions between parties without the need for an intermediary.

中文题干与选项 / Chinese translation

下列关于加密货币的陈述，哪项最准确？加密货币：

A. 表现出低波动性。

B. 对可发行的货币总量没有限制。

C. 允许交易双方无需中介即可进行交易。

答案 / Answer: C

1. 原答案强调，加密货币能够让交易双方无需银行等中介即可进行交易，因此C最准确。

The source emphasizes that cryptocurrencies can allow parties to transact without an intermediary such as a bank, making C the most accurate statement.

2. A错误，因为此类资产可能有很高价格波动；B把发行上限一概否定，原答案指出部分加密货币有发行限制。

A is incorrect because such assets can have high price volatility. B incorrectly denies supply limits universally; the source notes that some cryptocurrencies impose limits.

易错点 / Common trap

去中介不等于低风险；不要把某一种代币的发行规则推广到所有加密货币。

Disintermediation does not mean low risk; do not generalize one token's issuance rules to all cryptocurrencies.

资料提示 / Source note

跨科目提示：本题列在原题库数量分析部分，但答案页页脚标注为Alternative Investments（另类投资）；本讲义按金融科技知识点保留。

Cross-topic note: the question appears in the supplied Quantitative Methods section, but its answer page is labeled Alternative Investments. It is retained here under fintech.

Q77 资产代币化 / Asset tokenization

核心例题 / Core example • 原题 PDF p.130 / 章节 p.16 • 答案 PDF p.77

77. The process of representing ownership rights to physical assets on a distributed ledger is referred to as:
    A. tokenization.
    B. initial coin offering.
    C. consensus mechanism.

中文题干与选项 / Chinese translation

在分布式账本上表示实物资产所有权的过程称为：

A. 代币化。

B. 首次代币发行。

C. 共识机制。

答案 / Answer: A

1. 把实物资产所有权映射为账本上的数字代币，正是资产代币化，因此选A。

Representing ownership rights to physical assets as digital tokens on a ledger is asset tokenization, so A is correct.

2. 首次代币发行描述发行代币进行融资；共识机制描述网络如何对记录达成一致，均不等于把所有权数字化表示。

An initial coin offering concerns raising funds through token issuance, while a consensus mechanism concerns agreement over records. Neither defines the representation of ownership rights.

易错点 / Common trap

题干关键词是ownership rights，而不是融资或网络验证。

The key words are ownership rights, not fundraising or network validation.

资料提示 / Source note

跨科目提示：本题列在原题库数量分析部分，但答案页页脚标注为Alternative Investments（另类投资）；本讲义按金融科技知识点保留。

Cross-topic note: this question appears in the supplied Quantitative Methods section, but its answer page is labeled Alternative Investments. It is retained here under fintech.

Q79 代币化与所有权转移 / Tokenization and ownership transfer

巩固练习 / Reinforcement • 原题 PDF p.130 / 章节 p.16 • 答案 PDF p.79

79. Which of the following statements about distributed ledger technology is *most* accurate?
    A. Bitcoin uses a permissioned network.
    B. Miners execute smart contracts in the blockchain.
    C. Tokenization can streamline the transfer of ownership of physical assets.

中文题干与选项 / Chinese translation

关于分布式账本技术，下列哪项最准确?

A. 比特币采用许可制网络。

B. 矿工在区块链中执行智能合约。

C. 代币化可以简化实物资产所有权的转移。

答案 / Answer: C

1. 资产权利在账本上被表示为代币后，可以支持更便捷的权利转移；这正是原答案选C的依据。

Representing asset rights as tokens on a ledger can facilitate transfers of those rights, which is the source's reason for choosing C.

2. A将比特币网络误称为许可制网络。B把智能合约执行泛化归于矿工，不是适用于分布式账本技术的一般准确描述。

A incorrectly labels the Bitcoin network as permissioned. B broadly assigns smart-contract execution to miners, which is not a generally accurate description of distributed ledger technology.

易错点 / Common trap

区分“资产权利表示”“共识/验证”“合约执行”三个不同功能，不要混为一谈。

Distinguish representing asset rights, consensus or validation, and contract execution; they are different functions.

资料提示 / Source note

跨科目提示：本题列在原题库数量分析部分，但答案页页脚标注为Alternative Investments（另类投资）；本讲义按金融科技知识点保留。

Cross-topic note: this question appears in the supplied Quantitative Methods section, but its answer page is labeled Alternative Investments. It is retained here under fintech.

附录 A 公式速查

Appendix A • Formula reference

收益率与货币时间价值 / Returns and time value

HPR = (P1 - P0 + D) / P0

持有期收益包含价格变化和现金分红。

Holding-period return includes the price change and cash dividend.

对应原题 / Originals: Q3, Q9

Rannual = (1 + Rperiod)^(1/T) - 1

T 为持有期的年数；16 个月时指数为 12/16。

T is the holding period in years; for 16 months the exponent is 12/16.

对应原题 / Originals: Q6, Q8

TWRannual = [∏(1 + Rt)]^(1/N) - 1

先按外部现金流切分，再连乘。

Split at external cash flows, then geometrically link subperiod returns.

对应原题 / Originals: Q37, Q91

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0 = Σ CFt / (1 + MWR)^t

MWR 是真实投入与回收现金流的 IRR；关注时点和正负号。

MWR is the IRR of actual contributions and distributions; keep timing and signs consistent.

对应原题 / Originals: Q26, Q59

EAR = (1 + rnom/m)^m - 1; rc = ln(1 + EAR)

比较不同复利频率先统一到有效年收益率。

Convert compounding conventions to an effective annual rate before comparing.

对应原题 / Originals: Q54, Q74

PV = FV / (1 + rnom/m)^(mT)

利率和期数必须按同一复利频率计量。

Match the rate per period to the number of compounding periods.

对应原题 / Originals: Q78

PVordinary = PMT × [1 - (1+r)^(-n)] / r

普通年金公式得到第一笔付款前一期的价值。

An ordinary-annuity formula gives value one period before the first payment.

对应原题 / Originals: Q13, Q67

PVdue = PVordinary × (1+r)

期初年金的每笔付款比普通年金早一期。

Every annuity-due payment occurs one period earlier.

对应原题 / Originals: Q31, Q66

500,000 = PMT × Σ(t=0 to 9) 1.04^(15-t)

Q63 共 10 笔付款，发生在 t=0 至 t=9。

Q63 has 10 payments at t=0 through t=9.

对应原题 / Originals: Q63

描述统计、概率与组合 / Statistics, probability and portfolios

Arithmetic mean = Σxi/n; Harmonic mean = n/Σ(1/xi)

等额资金买入、求平均单位成本用调和平均；xi 要为正。

Use a harmonic mean for average unit cost under equal cash investments; xi must be positive.

对应原题 / Originals: Q39, Q55

Percentile position = (n+1)p; IQR = Q3 - Q1

p 为分位概率；边界及插值按题目所用约定。

p is the quantile probability; follow the source convention for boundaries and interpolation.

对应原题 / Originals: Q35, Q38, Q43

s = √[Σ(xi - x̄)^2/(n-1)]; MAD = Σ|xi - x̄|/n

样本标准差使用 n-1；平均绝对偏差使用 n。

Sample standard deviation uses n-1; mean absolute deviation uses n.

对应原题 / Originals: Q58

Target downside = √[Σ min(0, Ri - B)^2/(n-1)]

本题库使用样本目标下行偏差口径；分母不是低于目标的期数。

This bank uses the sample target-downside convention; the denominator is not the count of below-target periods.

对应原题 / Originals: Q45, Q70

CV = s/x̄; ρXY = Cov(X,Y)/(σXσY)

CV 是每单位平均收益的风险；相关系数只衡量线性关系。

CV measures risk per unit of mean return; correlation measures linear association.

对应原题 / Originals: Q46, Q48-Q50, Q81

E(X) = Σpixi; Var(X) = Σpi[xi - E(X)]²

概率分布计算使用概率权重，不再除以 n-1。

Use probability weights for a specified distribution, without dividing by n-1.

对应原题 / Originals: Q42, Q52

$E(D|F) = \Sigma P(Dj|F)Dj$

已知有利情景时，只用该情景内的条件概率。

When the favorable scenario is given, use probabilities conditional on that scenario.

对应原题 / Originals: Q53

$Cov(X,Y) = \Sigma\Sigma pij(xi - \mu X)(yj - \mu Y)$

收益率若用百分数单位输入，结果单位为百分比平方。

Inputs in percentage points produce covariance in percentage-points squared.

对应原题 / Originals: Q57

$\sigma p^2 = w1^2\sigma1^2 + w2^2\sigma2^2 + 2w1w2\rho12\sigma1\sigma2$

先算组合方差，再开平方；相关为零不等于组合风险为零。

Compute variance first, then take its square root; zero correlation does not eliminate risk.

对应原题 / Originals: Q62

Safety-first ratio = $[E(Rp) - RL]/\sigma p$

在正态收益假设下，最大化该比率可最小化跌破门槛概率。

With normal returns, maximizing the ratio minimizes the probability of falling below the threshold.

对应原题 / Originals: Q19, Q32

抽样、检验与回归 / Sampling, tests and regression

$SE(\bar{x}) = \sigma/\sqrt{n}$ (or estimated by $s/\sqrt{n}$)

标准误衡量统计量的抽样波动；抽样误差是一次样本与总体参数的差。

Standard error measures sampling variability; sampling error is one estimate’s deviation from the parameter.

对应原题 / Originals: Q1, Q40

Confidence interval = Estimate $\pm$ Critical value $\times$ SE

置信区间使用标准误，不能用不可知的抽样误差代入。

A confidence interval uses standard error, not the unknown realized sampling error.

对应原题 / Originals: Q1

Power = $1 - \beta$; Type I probability = α

拒真为 I 类；不拒假为 II 类。

Rejecting a true null is Type I; failing to reject a false null is Type II.

对应原题 / Originals: Q36, Q64, Q93

$t_{paired} = (\bar{d} - d0)/(sd/\sqrt{n})$; $df = n-1$

配对检验先把每一对观测变成一个差值。

For a paired test, first turn each pair into one difference.

对应原题 / Originals: Q5, Q34

$t_{correlation} = r\sqrt{(n-2)}/\sqrt{(1-r^2)}$; df = n-2

用于检验总体线性相关系数为零，须满足检验条件。

Tests zero population correlation under the relevant parametric assumptions.

对应原题 / Originals: Q17

$r_s = 1 - 6\Sigma d_i^2/[n(n^2-1)]$

此简式适用于无并列名次；Q83 的 $\Sigma d_i^2=12$。

This shortcut assumes no tied ranks; Q83 has $\Sigma d_i^2=12$.

对应原题 / Originals: Q83

Chi-square independence df = (rows-1)(columns-1)

列联表独立性检验自由度不是行数加列数。

Contingency-table degrees of freedom are not the sum of row and column counts.

对应原题 / Originals: Q16

$\hat{y} = b_0 + b_1x$; $e_i = y_i - \hat{y}_i$

Y 是被解释变量；残差使用样本拟合值。

Y is the explained variable; a residual uses the sample fitted value.

对应原题 / Originals: Q2, Q68, Q84

$SST = SSR + SSE$; $R^2 = SSR/SST$; $SEE = \sqrt{[SSE/(n-2)]}$

含截距的一元 OLS：SSE 为残差平方和；R^2 是比例，SEE 有 Y 的单位。

For simple OLS with an intercept, SSE is residual variation; R^2 is a fraction and SEE has Y’ s units.

对应原题 / Originals: Q23, Q44, Q76

Prediction interval = $\hat{y}_f \pm t_{critical} \times s_f$

先预测 Y，再用题目给出的预测标准误 sf 构造区间。

Predict Y first, then construct the interval using the supplied forecast standard error sf.

对应原题 / Originals: Q2, Q65

自由度一眼区分 / Degrees of freedom at a glance

场景 / Setting	自由度 / df
单一样本均值或配对差值 / One-sample or paired mean	n - 1
两独立样本、共同方差 / Independent means, equal variances	n1 + n2 - 2
相关系数或一元回归残差 / Correlation or simple-regression residuals	n - 2
列联表独立性 / Contingency-table independence	(r - 1)(c - 1)

附录 B 全部答案与考点索引

Appendix B • Answer and topic index

先完成题目再核对。星号表示必须结合资料提示阅读：Q19 的题干缺数据；Q74 的原选项数值有误。

Attempt each question before checking. An asterisk requires the source note: Q19 lacks question data, and Q74 has incorrect printed option values.

原题 Q	答案 Key	单元 Unit	考点 / Concept
Q01	B	06	置信区间的构成 Components of a confidence interval
Q02	B	08	回归预测区间 Regression prediction interval
Q03	C	01	持有期收益 Holding-period return
Q04	C	07	参数检验与非参数检验 Parametric versus nonparametric tests
Q05	A	07	均值差的检验统计量 Statistic for a mean-difference test
Q06	B	01	非整年收益年化 Annualizing a partial-year holding period
Q07	C	09	加密货币的交易特征 Transaction characteristics of cryptocurrencies
Q08	A	01	复合年增长率 Compound annual growth rate
Q09	A	01	由总收益反求卖出价 Solving for the sale price from total return
Q10	C	01	相对最佳替代品的机会成本 Opportunity cost relative to the best alternative
Q11	A	09	算法交易与低延迟 Algorithmic trading and low latency
Q12	C	09	过度拟合与拟合不足 Overfitting and underfitting
Q13	B	02	首笔在第3年的递延年金 Deferred annuity starting in Year 3
Q14	C	09	传统顾问的相对优势 Relative advantage of traditional financial advisers
Q15	C	07	单尾备择假设 One-sided alternative hypothesis
Q16	B	07	卡方独立性检验自由度 Degrees of freedom for a chi-square independence test
Q17	B	07	零相关系数的t检验 t-test of zero correlation
Q18	B	03	负偏分布的中心位置 Center of a negatively skewed distribution
Q19	C*	05	安全第一比率与缺失表格 Safety-first ratios with an incomplete source table

原题 Q	答案 Key	单元 Unit	考点 / Concept
Q20	C	08	线性化变换 Transformations to achieve linearity
Q21	A	03	正偏分布的中心位置 Center of a positively skewed distribution
Q22	C	09	金融科技的定义 Definition of fintech
Q23	B	08	估计标准误计算 Calculating the standard error of the estimate
Q24	A	06	Bootstrap识别 Identifying bootstrap sampling
Q25	C	01	名义无风险利率 Nominal risk-free rate
Q26	C	01	资金加权收益与股息时间 Money-weighted return and dividend timing
Q27	A	08	简单回归的F检验 F-test in simple regression
Q28	B	04	概率树的用途 Use of probability trees
Q29	C	09	用自然语言处理识别情绪 Using NLP to detect sentiment
Q30	B	08	相关性的图形检查 Graphical inspection of correlation
Q31	C	02	收款方案现值比较 Comparing the present values of payment alternatives
Q32	A	05	安全第一准则的风险目标 Risk objective of the safety-first criterion
Q33	C	06	概率抽样与代表性 Probability sampling and representativeness
Q34	B	07	配对t检验自由度 Degrees of freedom for a paired t-test
Q35	B	03	四分位距 Interquartile range
Q36	B	07	检验功效 Power of a test
Q37	B	01	含新增投入的时间加权收益 Time-weighted return with additional investment
Q38	B	03	分位组与边界 Quantile groups and boundaries
Q39	B	03	三种平均数的大小关系 Ordering three means
Q40	C	06	抽样误差定义 Definition of sampling error
Q41	B	06	中心极限定理与一致性 The central limit theorem and consistency
Q42	A	04	等概率情景的方差 Variance of equally likely outcomes

原题 Q	答案 Key	单元 Unit	考点 / Concept
Q43	A	03	偶数样本的中位数 Median of an even-sized sample
Q44	C	08	估计标准误的含义 Interpreting the standard error of the estimate
Q45	B	03	样本目标下行偏差 Sample target downside deviation
Q46	C	03	变异系数与夏普比率 Coefficient of variation versus Sharpe ratio
Q47	B	03	排序后的中位数 Median after sorting
Q48	C	03	相关系数的含义 Meaning of correlation
Q49	A	03	协方差固定时的相关系数 Correlation with fixed covariance
Q50	A	03	相关系数只衡量线性联系 Correlation measures linear association
Q51	A	06	Bootstrap标准误 Bootstrap standard error
Q52	A	04	非等概率情景的标准差 Standard deviation with unequal probabilities
Q53	C	04	有利情景下的条件期望 Conditional expectation in the favorable scenario
Q54	B	01	季度复利与连续复利等价 Equivalent quarterly and continuous compounding
Q55	A	03	等额投入的平均成本 Average cost under equal cash investments
Q56	B	08	对数模型命名 Naming logarithmic models
Q57	C	04	由联合概率计算协方差 Covariance from joint probabilities
Q58	A	03	平均绝对离差与样本标准差 Mean absolute deviation and sample standard deviation
Q59	A	01	中途市价不等于现金流 An interim market price is not a cash flow
Q60	B	01	资金加权收益的含义 Meaning of the money-weighted return
Q61	C	01	资金流时机对收益率的影响 Effect of cash-flow timing on measured returns
Q62	B	05	零相关两资产组合风险 Risk of two uncorrelated assets
Q63	A	02	先付投入与远期目标 Contributions beginning today for a later target
Q64	C	07	功效与第二类错误 Power and Type II error
Q65	C	08	预测区间宽度 Width of a prediction interval

原题 Q	答案 Key	单元 Unit	考点 / Concept
Q66	B	02	递延学费的现值 Present value of deferred tuition payments
Q67	B	02	首笔在第5年的递延年金 Deferred annuity starting in Year 5
Q68	B	08	残差与总体误差 Residual versus population error
Q69	C	01	实际消费的时间偏好 Time preference for real consumption
Q70	A	03	目标等于均值时的半标准差 Semideviation when the target equals the mean
Q71	A	03	截尾平均数 Trimmed mean
Q72	C	05	正态对数收益与对数正态价格 Normal log returns and lognormal prices
Q73	B	06	非概率抽样 Non-probability sampling
Q74	C*	01	变动报价下的季度复利 Quarterly compounding with changing stated rates
Q75	B	05	对数正态分布的形状 Shape of a lognormal distribution
Q76	B	08	总平方和 Total sum of squares
Q77	A	09	资产代币化 Asset tokenization
Q78	A	02	半年复利下的单笔现值 Single present value with semiannual compounding
Q79	C	09	代币化与所有权转移 Tokenization and ownership transfer
Q80	A	08	回归的常规分布假设 Conventional distributional assumptions in regression
Q81	A	03	变异系数的方向 Direction of the coefficient of variation
Q82	A	07	改变假设均值差 Changing the hypothesized mean difference
Q83	A	07	Spearman等级相关 Spearman rank correlation
Q84	B	08	被解释变量的识别 Identifying the explained variable
Q85	A	03	负偏收益的经济含义 Economic meaning of negative skew
Q86	B	08	残差图诊断 Residual-plot diagnostics
Q87	A	06	整群抽样的应用 Applications of cluster sampling
Q88	B	06	非正态总体的样本均值 Sample means from a non-normal population

原题 Q	答案 Key	单元 Unit	考点 / Concept
Q89	A	03	由长尾识别偏度 Identifying skew from the longer tail
Q90	B	07	排名数据与非参数检验 Rank data and nonparametric tests
Q91	A	01	客户控制资金流时评价经理 Evaluating a manager when the client controls cash flows
Q92	C	06	分层随机抽样 Stratified random sampling
Q93	C	07	第二类错误 Type II error

附录 C 原文勘误与口径说明

Appendix C • Source issues and conventions

下列说明保留原文证据，并把原答案与本讲义的计算或限定条件区分开。小节中的题目旁也有相同提示。

The notes below preserve the source evidence and distinguish the supplied answer from the guide’ s calculation or qualifications. The same notes appear beside the relevant questions.

Q19 安全第一比率与缺失表格 / Safety-first ratios with an incomplete source table

原题表格的标题和列错排；标准差为“待补充 / Missing information”。本讲义未把答案中出现的数据冒充为完整原题，答案 C 以答案文件所列数据为条件。原答案也把题目美元符号写成欧元，比例计算不受影响。

The original table has malformed headers and columns; standard deviations are Missing information. This guide does not present answer-source figures as if they appeared in the original question. C is conditional on those figures. The answer also switches the dollar symbol to euros, which does not affect the ratio.

Q74 变动报价下的季度复利 / Quarterly compounding with changing stated rates

原题选项存在排印问题：原文A/B/C分别为697,200、697,335、697,455。已保留原文；原答案C应理解为意图对应约97,455欧元，而非认可697,455。

The printed choices contain an apparent typographical error: A/B/C are 697,200, 697,335, and 697,455. They are preserved; source answer C is understood as the intended choice for about €97,455, not as validation of 697,455.

Q53 有利情景下的条件期望 / Conditional expectation in the favorable scenario

答案页文字说明中曾把第一个股利写成 $2.50，但原题及答案页公式均为 $2.00。本讲义按原题 $2.00 计算，并保留此勘误说明。

The answer prose once labels the first dividend $2.50, but the original question and answer formulas use $2.00. This guide uses the original $2.00 and records the typo.

Q55 等额投入的平均成本 / Average cost under equal cash investments

原表数值列标题误写为“Term Deposit 1（定期存款 1）”；题干与答案均按基金每股买入价处理。保留原数值，不补造投入金额。

The original numeric-column header says “Term Deposit 1”, but the stem and solution treat the amounts as fund purchase prices per unit. Values are preserved and no investment amount is invented.

Q42 等概率情景的方差 / Variance of equally likely outcomes

原题把方差单位写为 $；正确单位应为 $²。原答案页计算处使用 $²，数值答案 A 不变。

The stem labels variance in dollars; the correct unit is dollars squared. The answer's calculation uses squared-dollar units, and the numerical answer remains A.

Q04 参数检验与非参数检验 / Parametric versus nonparametric tests

这不是“50个样本一定足够”的定理。极强偏度或厚尾时，样本均值的正态近似仍可能较差；实际使用需检查分布与检验条件。

This is not a theorem that 50 observations are always sufficient. With severe skewness or heavy tails, the normal approximation for the mean may still be poor, so practical use requires checking the distribution and test conditions.

Q05 均值差的检验统计量 / Statistic for a mean-difference test

原解析用配对样本公式作说明，题干本身未明确写出配对关系；不能仅凭这道题据此认定所有均值差检验都用n - 1自由度。

The supplied explanation illustrates a paired-sample formula, but the stem does not explicitly specify pairing. Do not infer that all mean-difference tests have n - 1 degrees of freedom.

Q07 加密货币的交易特征 / Transaction characteristics of cryptocurrencies

跨科目提示：本题列在原题库数量分析部分，但答案页页脚标注为Alternative Investments（另类投资）；本讲义按金融科技知识点保留。

Cross-topic note: the question appears in the supplied Quantitative Methods section, but its answer page is labeled Alternative Investments. It is retained here under fintech.

Q08 复合年增长率 / Compound annual growth rate

原答案对选项B的说明写成ln(1.125) - 1 = 0.11778，算式与数值不一致；实际ln(1.125)约为0.11778，不应再减1。正确答案A不受影响。

The source explanation for B writes ln(1.125) - 1 = 0.11778, which is inconsistent. In fact ln(1.125) is approximately 0.11778, without subtracting 1. The correct choice A is unaffected.

Q20 线性化变换 / Transformations to achieve linearity

原解析用“不相关”直接推出“独立”，一般并不成立；独立通常能推出不相关（相关矩存在时），反向需要额外条件。此处不影响C答案。

The supplied explanation directly infers independence from zero correlation, which is not generally valid. Independence implies zero correlation when the relevant moments exist, but the converse needs additional conditions. Answer C is unaffected.

Q33 概率抽样与代表性 / Probability sampling and representativeness

原解析把概率抽样概括成每个成员入选概率相同，表述过窄。更一般的要求是入选概率已知且非零，并不一定相等；C答案不变。

The supplied explanation describes all probability sampling as giving members equal selection probabilities, which is too narrow. The general requirement is known, nonzero inclusion probabilities, which need not be equal. Answer C is unchanged.

Q38 分位组与边界 / Quantile groups and boundaries

原题印作“19h”，按答案中的“19th”理解为第 19 个；分位边界遵循本资料的约定。

The original prints “19h”; the answer identifies it as the 19th observation. Quantile boundaries follow the supplied convention.

Q39 三种平均数的大小关系 / Ordering three means

原题写“非负”而非“严格为正”，调和平均数的通常公式需要所有观测大于零。答案 B 为原解析的预期答案；此处补充适用条件，不改原题。

The stem says non-negative rather than strictly positive, although the usual harmonic-mean formula requires positive observations. B is the intended supplied answer; this note states the domain without changing the question.

Q41 中心极限定理与一致性 / The central limit theorem and consistency

严格概念上，一致性通常直接由大数定律说明。这里保留原题B，同时补足其适用的常规条件。

Strictly, consistency is most directly established by the law of large numbers. B is retained as the supplied answer with its usual conditions made explicit.

Q45 样本目标下行偏差 / Sample target downside deviation

本讲义按用户提供答案采用样本口径 n-1；实际应用中也有以 n 为分母的下行偏差定义，须先确认约定。

This guide follows the supplied answer's sample convention n-1. Some applications define downside deviation with n, so identify the convention first.

Q64 功效与第二类错误 / Power and Type II error

原解析把p值说成与第一类错误概率同义，这不准确。alpha是预先设定的错误率阈值，p值根据观测数据在H0下计算；两者并非同一数值。C答案正确。

The supplied explanation equates the p-value with the Type I error probability, which is inaccurate. Alpha is a prespecified error-rate threshold; the p-value is calculated from the observed data under H0. They are not the same quantity. Answer C is correct.

Q77 资产代币化 / Asset tokenization

跨科目提示：本题列在原题库数量分析部分，但答案页页脚标注为Alternative Investments（另类投资）；本讲义按金融科技知识点保留。

Cross-topic note: this question appears in the supplied Quantitative Methods section, but its answer page is labeled Alternative Investments. It is retained here under fintech.

Q78 半年复利下的单笔现值 / Single present value with semiannual compounding

原答案的计算器输入行写作I/Y = 0.025。若计算器利率栏按百分数输入，采用每半年一期、N = 30时，应输入2.5，表示2.5%；不要照抄为0.025%。原答案的折现公式和选项A正确。

The source calculator line gives I/Y = 0.025. If the calculator rate field uses percentage input, with each period representing half a year and N = 30, enter 2.5 for 2.5%, not 0.025%. The source discounting formula and choice A are correct.

Q79 代币化与所有权转移 / Tokenization and ownership transfer

跨科目提示：本题列在原题库数量分析部分，但答案页页脚标注为Alternative Investments（另类投资）；本讲义按金融科技知识点保留。

Cross-topic note: this question appears in the supplied Quantitative Methods section, but its answer page is labeled Alternative Investments. It is retained here under fintech.

Q80 回归的常规分布假设 / Conventional distributional assumptions in regression

原题使用residuals一词。本讲义保留题干与A答案，并区分理论误差项与估计后的样本残差。

The original stem uses the word residuals. The stem and answer A are retained while distinguishing theoretical errors from estimated sample residuals.